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Bachelor Thesis

Fiscal Balance and Economic Growth: A Relationship Conditioned by Sovereign Risk

An Empirical Analysis

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Abstract

This thesis examines the dynamic effects of government spending, taxes and sovereign risk in Argentina, a country characterized by persistent fiscal imbalances. To capture the dynamic relationship, a Structural Vector Autoregressive (SVAR) model following Blanchard and Perotti ([2002](#)) augmented with the Riesgo País as a measure of sovereign risk is implemented. The results indicate that government spending shocks have an insignificant impact on output, while tax shocks stimulate economic growth. These findings highlight the importance of maintaining a fiscal balance in fiscally vulnerable economies like Argentina.

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Chapter 1 Introduction

“There are four kinds of countries:
developed, emerging, Japan... and
Argentina”

— Kuznets as cited in Czerwonko

([2025](#))

Fiscal deficits have long been at the center of public and academic discussions. Austerity policies are measures used to counteract the increasing weight caused by fiscal indiscipline. Ideally, if governments behaved according to economic theory, they would accumulate fiscal surpluses during periods of economic growth. This would, in turn, enable countries to incur deficits to stimulate demand during recessions. Consequently, such austerity measures, where governments either increase taxes and/or decrease spending, should not be common. However, this is often not the case. The natural consequence of government spending beyond their means is an increase in public debt. This becomes especially problematic when economic growth alone cannot decrease the burden of total debt, which can also be an impediment to further growth. As a consequence, investors could lose confidence in the government’s ability to repay the debt (Alesina, Favero & Giavazzi, [2019](#), pp. 1–2). Austerity policies are often necessary due to the accumulation of past mistakes by policymakers. This is specifically important for developing countries, where trust can be broken more quickly and with far higher costs than in developed countries. The consequences of this risk, which translates into distrust, can be multiple.

Argentina is a notable example of the economic consequences of fiscal indiscipline. In the

early years of the 20th century, it belonged to the richest countries in the world. Nevertheless, due to its difficulties in managing its fiscal balance, the country started to diverge from the rich world following a path characterized by high inflation rates, stagnation and defaults. The economic history of Argentina illustrates how persistent fiscal instability and credibility issues can destabilize the economic system, giving a concrete motivation for this research. Consequently, this thesis aims to answer the following research question "How do government spending and tax revenue shocks interact with sovereign risk in shaping real output in Argentina?". To answer this question, a Structural Autoregressive model (SVAR) following the method of Blanchard and Perotti (2002) is expanded by including a sovereign risk variable. This study thus aims to analyze the short term effects of fiscal shocks and how sovereign risk conditions these variables.

The structure of this thesis is as follows. The literature review and a summary of Argentina's recent economic history are presented in Chapter 2. First, the dynamics of fiscal policy are reviewed, focusing on the role that government spending, taxes and debt can have on economic growth. Second, the impact of sovereign risk on the economy is explored. In particular, what are the reasons for the government's debt repayment and what are the empirical determinants of the sovereign risk as well as its economic consequences. Finally, the chapter reviews Argentina's recent economic history. Chapter 3 delves into the model and data used in this thesis. First an introduction to Vector Autoregressive model (VAR) and Structural Vector Autoregressive (SVAR) model is done. Subsequently, the model presented in this study and the ideal identification strategy is discussed. Specifically, a comparison between the Recursively Identified models and the strategy presented in Blanchard and Perotti (2002) is made, with the latter identified as the more suitable option. Finally the chapter concludes by presenting the data, identifying the optimal lag length and outlining the estimation strategy used for the exogenous parameters. Chapter 4 presents the empirical results. It begins by looking at the contemporaneous coefficients, followed by the exploration of the effects of the different shocks. Robustness checks are then done and finally the results are discussed. Finally, Chapter 5 concludes the thesis and makes some reflections on the policy implications.

Chapter 2 Literature Review and the Economic Context

"The effects of government debt and deficits on the economy are not obvious from either economic theory or statistical evidence"

— Seater ([1993](#), p. 142)

Since austerity measures may become necessary due to fiscal indiscipline of governments, it is important to understand the role of government spending, debt and taxes on the overall economy. For example, if economic costs of reducing government spending outweigh the benefits, it would be counterproductive to decide to implement such policy. Consequently, the effects of government spending, taxes and sovereign risk on growth are examined in the following chapter.

2.1 Government Spending and Growth

2.1.1 Crowding Dynamics of Fiscal Policy

The effects of government spending has been studied for a long time. Nevertheless, neither theoretically nor empirically is there a clear consensus (Furceri & Sousa, [2011](#), p. 5167). In the Keynesian framework, an increase in government spending has a multiplier effect on aggregate demand and overall production (Hemming, Kell, & Mahfouz, [2002](#), p. 4).

2.1. Government Spending and Growth

Specifically, an increase in government spending should increase aggregate demand, which consequently results in higher labor demand and therefore higher wages and employment. Finally, this would increase consumption. The multiplier effect is the combined result of the direct effect of the increase in government spending and the subsequent increase in private consumption (Hagedorn, Manovskii & Mitman, 2019, p. 1).

However, the simplest Keynesian model can be extended to incorporate crowding-out effects. As explained by Buiter (1977), this effect refers to "the displacement of private economic activity by public economic activity" (p. 9). In other words, government spending may reduce the size of the multiplier effect, though it remains generally positive. In Keynesian models, crowding-out can happen in three principal ways: decrease in investment, an appreciation of the exchange rate and price flexibility (Hemming et al., 2002, pp. 4–6). First, since investment is inversely related to interest rates, expansionary fiscal policy increases income and money demand, consequently also raising interest rate. Higher interest rates make some investments unprofitable and reduce the value of financial assets. If consumption depends partly on financial wealth, this can have an additional negative effect. At the same time, higher interest rates attract additional foreign capital, appreciating the exchange rate and reducing the trade balance. Finally, if prices are partially flexible in the short run, an expansionary policy may be less effective, as an increase in demand can raise prices (Hemming et al., 2002, pp. 4–6).

However, these are not the only possible channels through which crowding-out effects can occur. New classical models focus more on microeconomic foundations and thus find other possible crowding-out channels. For example, within the Ricardian Equivalence framework, an increase in the government deficit is seen as a future tax liability, which offsets the perceived positive effects of holding government debt. This argument was originally proposed by Barro (1974), who discussed the neutrality of government debt. Consequently, the consumers smooth their consumption over time based on government's debt policy (Hemming et al., 2002, p. 8). Empirically, however, the evidence supporting this view is weak. Yet, for example, Panizza and Presbitero (2014) studied a sample of OECD countries, which initially showed a negative correlation between debt and growth.

Nonetheless, once they corrected their econometric model for endogeneity, they found no statistically significant effect of debt on growth. Similarly, Hsieh and Lai (1994) studied the G7 countries, which are notably advanced economies, found that the relationship between government spending and per capita output growth is heterogeneous over time and countries, with no convincing evidence that higher spending increases income.

Other plausible explanations for the crowding-out effect include credibility issues, risk premia, and uncertainty. The increase in the debt burden makes economies more vulnerable to crises in international confidence and increases the danger of lower political independence (Elmendorf & Mankiw, 1999, p. 1632). For example, Fels, Froehlich, Blanchard, and Miller (1987) in their study on the German Fiscal consolidation policy in the 1980s, attribute the growth of the economy to the cut in government spending as it increased fiscal stability and predictability of the government. If this change is in line with the preferences of the private sector, then it might stimulate the private sector (Fels et al., 1987, p. 184). However, this psychological fear can be exaggerated and thus might require the government to inform the private market about the exact conditions of the public fiscal balance (Miller, Skidelsky & Weller, 1990, p. 309). This, however, works only if policymakers are trustworthy. The sovereign risk channel is further developed in Chapter 2.2.

In the long run, debt and fiscal deficits can hinder growth as excessive government spending can have adverse effects. This occurs when decreased public saving (through increased government spending and/or tax cuts) is not completely offset by an increase in private savings. The reduction in national savings, due to the higher marginal product of capital, increase interest rates. Over the long run, the reduction in investment causes a decrease in capital, and consequently, lower output and income (Elmendorf & Mankiw, 1999, p. 1629). Nevertheless, while this is an important factor, it is not the only one. Countries with large debt are more likely to face higher interest rates, making the debt more costly. In the short run, the monetary authority can reduce the rates through expansionary policies. However, this does not have a significant impact in the long run (Elmendorf & Mankiw, 1999, p. 1630). In extreme cases, if the fiscal authorities are not able to finance

their deficit with new bond sales and there is no willingness to adjust their fiscal balance, then the monetary authority is forced to create new money through seigniorage, generating inflation (Sargent & Wallace, 1984, p. 2). Another cost of government debt may come from the deadweight loss caused by the distortionary taxes that are required in the future to repay the debt (Elmendorf & Mankiw, 1999, p. 1631). In addition, the ability to take on government debt can weaken fiscal discipline among politicians and the public, particularly when the increase in spending does not have to be accompanied by higher taxes (Elmendorf & Mankiw, 1999, p. 1631). Interestingly, Butkiewicz and Yanikkaya (2011) found that in developing countries with ineffective governments, the effect of government consumption reduces growth.

Abbas and Christensen (2009, pp. 213–214) however, argue that debt can also have positive effects on growth, inflation and savings by creating and improving capital markets, specifically for developing countries. Specifically, a domestic debt market is fundamental for the development of the country’s national financial system. First, yields on government bonds can fundamentally work as a benchmark, which can develop the corporate bond market and stimulate more competition in the banking system. Second, having a domestic bond market offers a relatively secure market, which can discourage capital flight and transfer money from the informal market into the financial sector (Abbas & Christensen, 2009, pp. 213–214). Furthermore public debt can be used as a tool of economic management for the government. For example, it can enable the central bank to conduct indirect monetary policy and use the debt as collateral for interbank lending. This becomes especially beneficial when the alternative policies are much more restrictive (such as interest rate controls). Finally, by improving its financial systems, mobilizing savings and increasing political accountability, domestic debt markets become fundamental for the development of a country and to build the necessary credibility to access international markets (Abbas & Christensen, 2009, pp. 212–214). Therefore, they are critical of policies that promote zero net domestic financing policies in countries with low levels of debt and underdeveloped capital markets. Thus, a diminishing of the size of debt could actually have adverse effects. Empirically, Abbas and Christensen (2009) found that pub-

lic debt can have a positive impact on economic growth. However, they also found some evidence of a reversal of this beneficial effect. Specifically, once the threshold of 23% of debt to GDP is surpassed the burden of public debt might start hurting the growth of the country. Similarly, other studies focusing on developing countries have found that government spending can have an important positive effect. Specifically, if aimed towards agriculture, education and road building, it can strongly contribute to economic growth. This suggests that not all spending is productive and, consequently, the aim should be to reallocate resources toward the more productive sectors (Bose, Haque & Osborn, 2007; Fan & Rao, 2003).

Consequently, government spending and debt can have various effects that can be positive, negative or even neutral. Depending on the different levels, different results can appear. This strongly suggests a non-linear relationship. In fact, several studies show that once a certain threshold has been surpassed, growth starts to suffer (Cecchetti, Mohanty & Zampolli, 2011; Checherita-Westphal & Rother, 2012; Reinhart & Rogoff, 2010). For example, Checherita-Westphal and Rother (2012) found a threshold for which debt becomes a burden on growth which lies between 90% and 100% of GDP. Nevertheless, as Égert (2014, pp. 237–238) points out, there is no magic number at which debt becomes automatically much more costly. This strongly depends on the time dimension and country coverage. Similarly, government spending can also have non-linear effects. For example, Christie (2012) found that the benefits of government spending depend on the size of the government, which is measured as spending to GDP. Specifically, if developed countries surpass the threshold of 26% the effect shows to be negative. Similarly, the threshold for developing countries stands at 33% and if the value is lower, additional spending presents a positive effect. Overall, these findings suggest an important conclusion: if the reduction is done through a cut in productive spending, this could come at high long term costs, whereas if it is used to reduce unproductive spending, there could be short and long term growth benefits.

As discussed above, government spending and debt may affect the economy's growth rate. However, taxes can also influence economic growth because they can affect the

choice making process by the different actors in the economy (Myles, 2000, p. 141). Specifically, what really matters is the marginal tax rate as it defines how individuals behave respectively to an increase in wealth and how much they can retain (Myles, 2000, p. 158). According to Engen and Skinner (1996, pp. 618–619), there are five channels through which taxes can affect the growth rate. First they can discourage investments by imposing high nominal taxes, high capital gains taxes and/or limiting depreciation deductions. Second, taxes can reduce the incentives to work and thus decrease the labor supply growth (Engen & Skinner, 1996, p. 619). Third, taxes can discourage productivity growth by reducing the incentives to innovate and limiting the venture capital availability. Fourth, taxes can distort the capital allocation by reducing the marginal return to capital in highly taxed sectors compared to less taxed and productive ones. Lastly, taxation of labor can also affect the allocation of human capital as workers may not want to work in sectors which generate high social value but are also highly taxed (Engen & Skinner, 1996, p. 619). Empirically, the evidence is consistent indicating that structure of the tax system may be as important as the tax level (Myles, 2000, p. 164). For example, corporate taxes have been found to have a significant negative effect on economic growth. Lee and Gordon (2004) estimate that a 10% decrease in the corporate tax rate can increase growth rate by 1.1% annually. On the other hand, taxes on immovable property appear to not have a significant effect (Macek, 2015). Consequently, the following ranking of taxes based on their effect on growth is obtained: corporate taxes are found to be the most harmful, followed by personal income taxes, then by consumption taxes and finally property taxes (Johansson, Heady, Arnold, Brys & Vartia, 2008, p. 2).

2.2 Sovereign Risk

In this section, the role of the Sovereign Risk in the economy is explored. Sovereign Credit Ratings are defined by Reinhart (2002) as "summary measures of the likelihood that a country will default" (p. 151). This definition can be extended to other measures of Sovereign Risk such as the spread between bond yields or between Credit Default Swaps, as investors value how risky the debt of a country is. This section will continue now by

firstly exploring the determinants of risk and subsequently the economic consequences of risk and default.

2.2.1 Theoretical Foundations of Sovereign Debt Repayment

Unlike corporate debt, where a failure to make a payment triggers legal enforcement, the enforcement mechanisms for sovereign debt are weaker. This is true because of two important factors: first, most of a country's assets are on national territory and second due to limited foreign jurisdiction (Panizza, F. Sturzenegger & Zettelmeyer, 2009, pp. 652–654). Even if over the years countries have moved towards a restrictive immunity, the full repayment of the debt, in case of a failure, is more the exception than the rule (F. Sturzenegger & Zettelmeyer, 2007). Given a partially absent legal enforcement mechanism, the fact that countries still repay their debts suggests that there are other factors.

On the theoretical side, different reasons were found. For example, Eaton and Gersovitz (1981) show that the threat of being permanently excluded from international credit markets is strong enough to not default. On the one hand, since there are no strong legal enforcements, countries will want to borrow as much as possible and then default with no (or little) costs. On the other hand, lenders will refuse to lend more than what they consider appropriate and thus setting a debt ceiling. This ceiling depends on the perception of lenders of the borrower's disutility of defaulting. However, as Kletzer and Wright (2000, pp. 622–623) note, an indefinite exclusion punishes debtors, but also lenders as they are not able to lend their money again. Thus, they would prefer that the countries are able to borrow money again. However, if a renegotiation is anticipated, and thus the threat loses credibility, the enforcement of repayment will be much more difficult (Panizza et al., 2009, p. 660). Additionally, as Bulow and Rogoff (1988, p. 6) note, there are still other instruments that countries can use instead of debt, such as consumption-insurance contracts. Moreover, if there are at least two lenders, this means that also other lenders need to forego the possibility of gains from trade. Nevertheless, while this model comes with its own important caveats, it presents an important tool of punishment that a country could suffer.

From this theoretical findings, other models were developed. This can be divided into three groups: direct punishments, repercussions in the credit market and collateral consequences of defaults (Panizza et al., 2009, p. 661). For example, Bulow and Rogoff (1986) introduce as cost the loss of gains from trade, which can be a reason for repayment by the debtor. Second, Kletzer and Wright (2000) develop an intertemporal barter model in which there is no external enforcement and all threats are subject to continuous renegotiation. The punishment in case of default consists in a short-lived payment. If this payment is enough to make the lender profitable, then the relationship between the two parties resumes (Kletzer & Wright, 2000, p. 623). Third, for example Cole and Kehoe (1998), expanded reputation models such as the one of Eaton and Gersovitz (1981), by exploring the idea that if a government is untrustworthy in one field, it will spillover to others. Therefore, they bring forward the problem that one issue cannot just be analyzed as the consequences in one field, but also how it expands to others and the possible consequences that arise from that. Similarly, the model developed by Sandleris (2008) uses defaults as signals as it can negatively impact the beliefs on the country. Consequently, investments may be reduced which can have a significant cost on the economy.

2.2.2 Determinants of Sovereign Risk

Empirical research found a wide range of financial and macroeconomic factors that influence sovereign risk. As the theoretical studies showed, fiscal variables such as debt to GDP ratios and fiscal deficits are associated with higher risk (Hilscher & Nosbusch, 2010; Ramos-Francia & Rangel, 2011; Petrova, Papaioannou & Bellas, 2010). Macroeconomic stability indicators, such as inflation and movements in the exchange rate, with higher inflation increasing perceived risk. On the other hand, international reserves and currency appreciations tend to lower it (Hilscher & Nosbusch, 2010; Ramos-Francia & Rangel, 2011). Additionally, global and structural conditions such as trade volatility and global liquidity stress and risk affect negatively sovereign risk (Hilscher & Nosbusch, 2010; Petrova et al., 2010). For developing countries, US interest rate shocks influence country spreads, which later transmit their effects to the economy (Uribe & Yue, 2005). Finally,

also development related factors such as GDP per capita or other indicators of economic development and the country's default history influence credit ratings and consequently also its sovereign risk (Cantor & Packer, 1996).

2.2.3 Economic Consequences of Sovereign Risk and Default

After discussing the theoretical background and the empirical determinants of sovereign risk, it is essential to discuss the economic empirical consequences.

Chiu and Lee (2017) analyzed the effects of public debt on economic growth depending on the risk environment in which a country is situated. They found that, in a high risk environment, a higher level of debt has a direct impact on the economic growth. Moreover, when one country moves from a high risk to a low risk environment, the negative effect of high public debt becomes weaker. Interestingly, in a low risk environment, higher debt can even stimulate economic growth, whereas in a risky environment the contrary is true. Similarly, Chen, Chen, Chang and Yang (2015) found that changes in the credit rating affects a country's economic growth. An upgrade by one notch in the credit rating increases economic growth by 0.6%, whereas a downgrade decreases growth by 0.3%. Moreover, Neumeyer and Perri (2005) found that eliminating default risk substantially decreases the output volatility in developing countries. Additionally, it not only affects the real economy, but also stock markets. Specifically, only credit rating downgrades reduce the national equity market's performance. This suggests that, once they receive negative information, international portfolios tend to reweigh their exposure to riskier markets (Brooks, Faff, Hillier & Hillier, 2003).

As sovereign risk increases, its consequences propagate throughout the economy. Corsetti, Kuester, Meier and Mueller (2012) analyzes the consequences of an increase in sovereign risk on the private credit market and finds that, as sovereign risk increases, borrowing costs also increase unless expansionary monetary policy offsets this effect. Consequently, this spreads to the entire economy. In such risky environments, where monetary policy is constrained, restrictive fiscal policies may even have expansionary effects. In addition, there may be a self-fulfillment problem: if the private sector expects a weak economy, this

belief alone can raise risk premium and thus impact negatively the economy. Moreover, Corsetti et al. (2012) found that sovereign risk amplifies the effect of economic downturns.

As private credit becomes more expensive with an increase in sovereign risk, the amount of credit available in the private market can also decrease. Bocola (2016) studies the relationship of sovereign risk and financial intermediation and identifies two main channels where sovereign risk has an effect on banks: the liquidity channel and the risk channel. First, if a bank holds bonds that become riskier, the prospect of loss decreases their market value, lowering the bank's net worth and consequently decreasing its lending capacity. Second, even if banks are not yet financially constrained, news of an increase in sovereign bond, can lead banks to be more reluctant to hold private sector assets. As banks anticipate that a default would cause balance sheet losses and force them to sell assets at lower prices, they will perceive loans to firms as riskier. Consequently, they demand higher returns for holding these assets and reduce their exposure by deleveraging, which further decreases credit to the private sector.

Finally, Xu, Lv and Xie (2018) show that as the riskiness in a country increases so does the informality of its economy. Specifically, a decrease in country risk by 1% causes a fall by 0.1% in informality. This has high costs on the economy as there might be an inefficient allocation of resources and official economic indicators (such as unemployment) become unreliable.

2.3 Argentina's Unique Economic History

The adoption of Argentina's Constitution in 1853 coincided with the first wave of globalization and until 1913 it was one of the fastest growing emerging countries (Cavallo & Runde, 2020, p. 90). It even became one of the ten richest countries in the world (Spruk, 2019, p. 1). However, the country's government have been plagued by one big shortcoming: for over 200 years they were not able to balance the budget consistently (Cavallo & Runde, 2020, p. 458). This section does not aim to explore the whole economic history of Argentina, but rather to review its recent historical background and to give a context

to the model applied in this thesis.

2.3.1 Historical Background and Legacy of the 1990s

From the highly successful economic history, which ended with the beginning of the first World War, Argentina found itself with a mixed economy characterized by a capitalist sector without a market and a socialist sector without a plan and thus experiencing the worst of both parts (A. Sturzenegger, [1984](#), p. 76). In 1989 Argentina was suffering from extreme inflation rate reaching a value of 3'079.5% (Ferrer, [2022](#), p. 387). In the same year, Carlos Menem became president and was convinced that the liberalization of international trade, reduction in government spending and fiscal deficit, reform of the tax system and find an agreement with the IMF were necessary policies (Cavallo & Runde, [2020](#), pp. 317–318). The first step taken in 1991 was the implementation of the "Ley de Convertibilidad", which primarily established a new monetary system based on the peso and permitted a one to one exchange with the dollar (Cavallo & Runde, [2020](#), p. 320). This policy was highly effective in reducing inflation, as in the following year inflation decreased to 84% and falling further reaching 4% in 1995, when Menem was reelected. This also caused a significant increase in investments in the country (Cavallo & Runde, [2020](#), pp. 322–326). Meanwhile, in 1992, with the Brady deal, Argentina was able to rejoin the international capital markets (F. Sturzenegger & Zettelmeyer, [2007](#), p. 165). Contemporaneously, international trade was liberalized and government spending to GDP was reduced from 34.7% in 1989 to its minimum of 22.5% in 1993. (Cavallo & Runde, [2020](#), pp. 325–337), while also other important liberalizations measures were implemented (F. Sturzenegger & Zettelmeyer, [2007](#), p. 168). This big policy package resulted in an annual growth rate of 6.3% from 1990 to 1998. Comparatively, over the same time Brazil grew at an average yearly rate of 2.7%, whereas in Argentina from 1980 to 1990 the yearly growth rate was equal to 0.7% (Cavallo & Runde, [2020](#), pp. 325–337).

2.3.2 The 2001 Crisis and Macroeconomic Recovery

Things however changed. On the one hand, by 1997 the presidential elections of 1999 were in sight. Menem, who was running for a third mandate, that was later impeded

by the Supreme Court, and Duhalde, Governor of the Province of Buenos Aires, used government spending as a tool to win the presidential nomination for the Peronist party (Cavallo & Runde, 2020, pp. 341–344). On the other side, the 1998 Russian default and the 1999 devaluation of the Brazilian Real put the newly established exchange regime under pressure (F. Sturzenegger & Zettelmeyer, 2007, p. 168). In addition, the 1999 recession reduced tax revenue and the restructured debt that had been issued in the early years of the decade were approaching maturity, so new bonds had to be issued at higher interest rates (F. Sturzenegger & Zettelmeyer, 2007, p. 168). This whole situation, brought the country back to a fiscal deficit situation. Even though some efforts were made to return to a fiscally sound path, this was not enough. This created negative expectations and many thought that the country was following an unsustainable debt path due to its stagnant economy (F. Sturzenegger & Zettelmeyer, 2007, p. 170). Even though \$29 billion USD worth bonds were voluntarily rescheduled and additional financial support by the IMF was received, this was not enough to avoid the default on the 24th of December 2001 and the later devaluation of the peso. Through the devaluation of the currency, the country substituted its imports and consequently was able to recover employment (F. Sturzenegger & Zettelmeyer, 2007, pp. 182–186). Nevertheless, this caused the peso to depreciate to 3.8 pesos per dollar in July of 2002 (Cavallo & Runde, 2020, p. 410). Finally in 2005 76% of the bonds were exchanged for new ones (F. Sturzenegger & Zettelmeyer, 2007, p. 192). This crisis came at a high cost: Real GDP decreased by 4.4% in 2001 and by 10.9% in 2002 respectively (F. Sturzenegger & Zettelmeyer, 2007, p. 166), unemployment peaked in 2002 at almost 22% and half of the population was considered poor (Damill, Frenkel & Rapetti, 2015, p. 1).

2.3.3 The Post-Crisis Period and the Kirchner Era (2003 - 2015)

After being hit by such severe crisis, Argentina surprisingly recovered quickly, and from the beginning of 2002 to the third quarter of 2008, the economy grew at a yearly average pace of 8% (Damill et al., 2015, pp. 3-6). Although inflation was low in 2003, in the following years it accelerated again and reached 22% in 2008 (Cavallo & Runde, 2020,

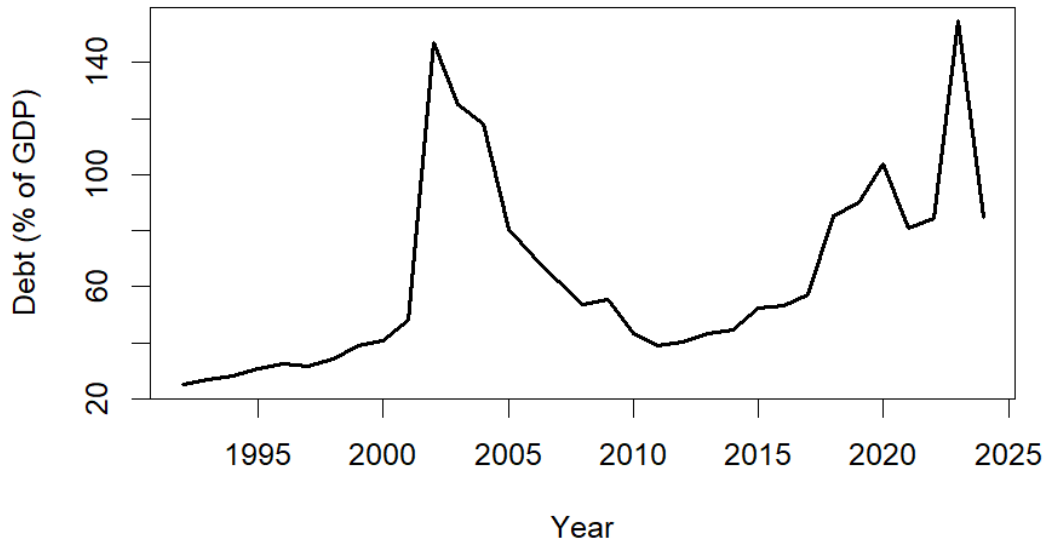


Figure 2.1: Debt to GDP in Argentina from 1992 to 2025. *Source:* IMF (2025)

p. 407). During this period, a stable and competitive real exchange rate was achieved through interventions in the foreign exchange rate market (Damill et al., 2015, p. 18), which undervalued the peso and thus stimulated the exports (Cavallo & Runde, 2020, p. 416). This was based on Néstor Kirchner's idea, that a undervalued peso would benefit Argentina's economic growth, but it required the printing of large amounts of money (Cavallo & Runde, 2020, pp. 435- 437). At the same time, taxes in exports were increased. This mix made it possible to achieve surpluses in both the fiscal balance and in the current account while also increasing the international funds (Damill et al., 2015, p. 6). To exit the economic crisis, emergency policies were carried out mainly in 2002-2003 under the interim President Duhalde. Specifically, to reduce inflation, which had increased from -1 % in 2001 to 41% one year later, he froze wages, pensions, and public utility rates, introduced taxes on exports and price controls (Cavallo & Runde, 2020, p. 406). After the collapse of the convertibility regime, also the central bank was modified. Some of these modifications are: the free floating of the exchange rate and inflation targeting, the obligation to announce and present inflation target and the evolution of the monetary aggregates and additional liberty in provide short term liquidity to illiquid commercial

2.3. Argentina's Unique Economic History

banks (Damill et al., 2015, pp. 16–17). In 2002, Néstor Kirchner was elected president, and in 2007 his wife, Cristina Kirchner, was elected as president (Cavallo & Runde, 2020, pp. 406–407). According to Cavallo and Runde (2020, p. 422) what distinguished mostly their policy after 2002 was the strong intervention of the state in trying to stabilize prices through taxes, quantitative regulations and controls of prices and the exchange rate. After 2005, as the fiscal surplus decreased, Néstor Kirchner had to find a new way to avoid a sudden appreciation of the exchange rate and therefore further increase the monetary which consequently increased also inflation (Cavallo & Runde, 2020, p. 426).

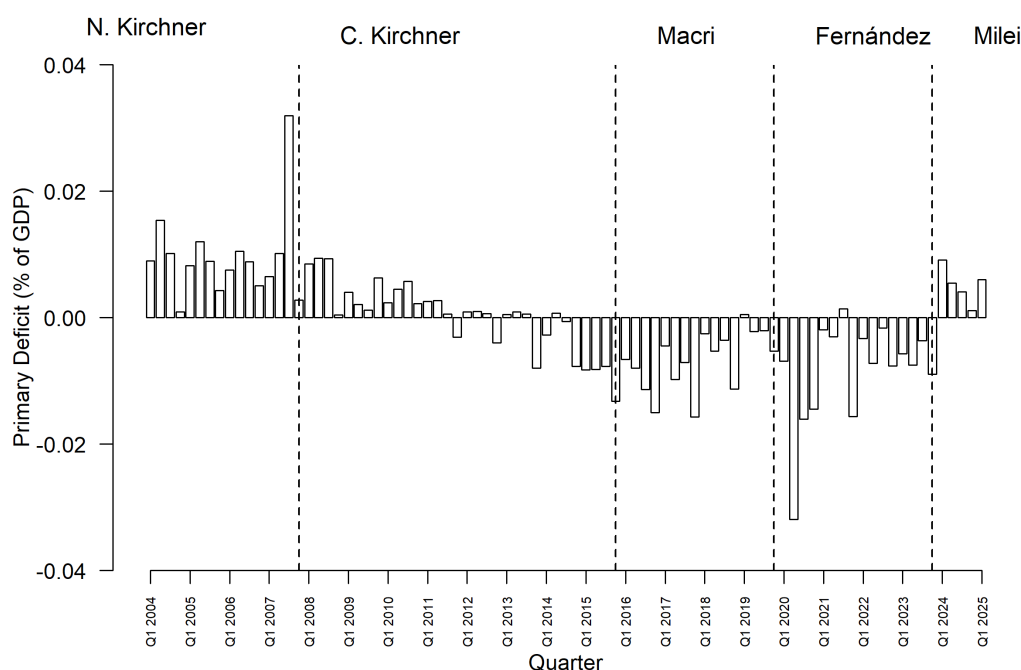


Figure 2.2: Argentina: Quarterly Primary Fiscal Balance. *Source:* Ministerio de Economía (n.d.)

In 2008, however, a combination of international and domestic factors complicated the situation. The 2008 financial crisis led investors to move from riskier to safer assets, while the nationalization of the pension system and the falsification of official statistics were sources of additional uncertainty which suddenly stopped the economic growth of Argentina (Damill et al., 2015, p. 8). Interestingly, according to Cavallo and Runde (2020, pp. 427–429), the nationalization of the pension system was basically used as a

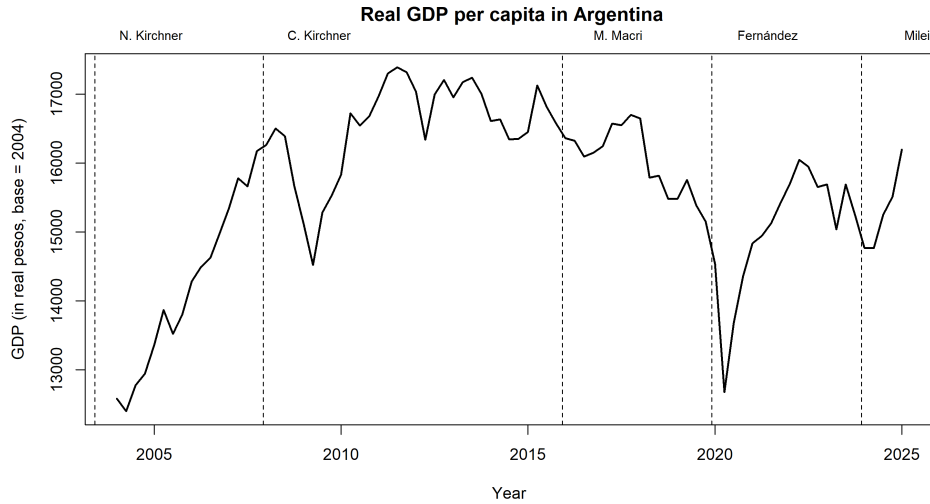


Figure 2.3: Real GDP per capita in Argentina. *Source:* INDEC (2025b).

tax on the population for additional government spending, and not as a benefit for (future) pensioners. Meanwhile, the fiscal balance began to deteriorate further (see Figure 2.2). Tax revenues rose from 25.2% of GDP to 37.2% in 2015, government spending increased even faster reaching in the same year 42.7% (Cavallo & Runde, 2020, p. 425). After the 2008 crisis, capital flight occurred also due to the realization that, even if nominally the peso was not overvalued, this was not longer true for the real exchange rate. This, in part, contributed to capital flight making it necessary to decrease the pace of money creation (Cavallo & Runde, 2020, p. 437). In 2011, the government under Cristina Kirchner imposed capital controls in an attempt to avoid a quick devaluation of the peso and consequently a further capital flight (Cavallo & Runde, 2020, p. 438). In the period after the crisis, 2009–2015, the inflation rate reached an average annual rate of 25.6%. The following years were characterized by stagflation (Cavallo & Runde, 2020, pp. 439–442). Following her reelection, she increased economic interventionism and implemented further controls on the exchange rate.

2.3.4 The Macri Presidency (2015 -2019)

On the 25th of October of 2015, Mauricio Macri was elected President of Argentina. According to F. Sturzenegger (2019, p. 340), who served as the governor of the Central Bank from December 2015 until his resignation in June 2018, Macri's election, a person

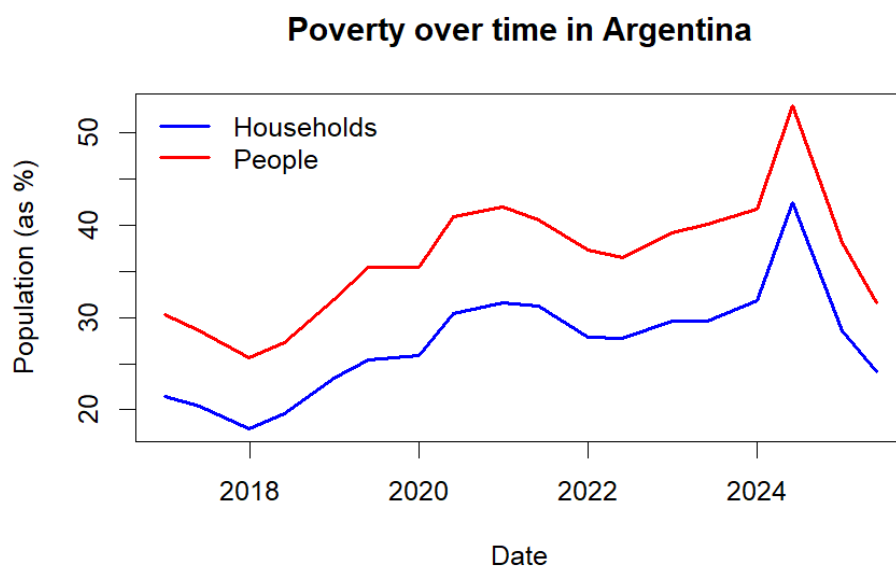


Figure 2.4: Poverty in Argentina *Source:* INDEC (2025b)

that came from the Argentine business elite, was seen as a relief preventing the country from moving towards a left wing authoritarian government after 14 years of Peronism. The inherited economy was in bad shape: Argentina had experienced four years of stagnation, while having high inflation, exchange rate controls and frozen utility prices (F. Sturzenegger, 2019, p. 341). The main economic and political strategy was based on the idea of gradualism: the fiscal deficit would have to be decreased gradually. This had two motivations. First, as can be seen in Figure 2.1, government debt was relatively low and the country could still borrow on international markets, meaning that a gradual cut would be more beneficial. Second, and this was the main reason, the government did not want to be labeled as right-wing. This would enable the President to preserve future political flexibility (F. Sturzenegger, 2019, pp. 345–346). Nevertheless, Macri's term ultimately ended up in failure: from 2016 to 2019 GDP grew by an average of -1.05% per year, inflation increased yearly on average by 38.2%, and the fiscal deficit, even if decreased, was not eliminated (F. Sturzenegger, 2019, p. 347). To F. Sturzenegger (2019, p. 341), this unsuccessful story is due to an expansionary fiscal policy that conflicted with the central bank's objectives, eroding monetary credibility and destabilizing the economy.

2.3.5 From Alberto Fernandez to Javier Milei (2019 - 2025)

On the 10th of December of 2019 Alberto Fernández took office and enacted the Solidarity and Productive Recovery Bill, which included fiscal reforms to increase the tax revenues through taxes on wealth, exports and foreign currency purchases combined with subsidies to channel these resources to the low income population to boost, in the short run, the economy (World Bank, 2022, pp. 2–42). During this period of strong economic imbalances, the COVID-19 pandemic hit the country in March 2020 and GDP fell by 9.9%. To support households and business, the government decided to implement a fiscal stimulus package that significantly increased the primary deficit (World Bank, 2022, p. 42). In the same year, the government tried to renegotiate the 68 billions USD in sovereign debt which were assessed to be unsustainable. Later the same year, the country had to restructure its sovereign debt in foreign currency (World Bank, 2022, pp. 2–3). Although additional taxes were implemented, this was not enough and the deficit had to be monetized (World Bank, 2022, p. 42). In 2021, Argentina achieved a 10.7% rise in GDP due partly to previous expansionary policies, elevated commodity prices, relaxed COVID-19 restrictions, and Brazil's (Argentina's primary trading partner) economic recovery. Nevertheless, because access to capital markets was limited, a large part of its deficit was still financed through monetization: in 2021 this accounted to 4.7% of the GDP (World Bank, 2022, pp. 42–43). In 2022 the country was hit by a severe drought that caused significant damage to the crops. The IMF (2023) described it as "the worst in six decades" (p. 33) causing a damage estimated about 20 billions USD relative to the previous year's harvest. The combination of these factors and the inability of the central bank to implement an effective monetary policy caused the inflation rate to rise (as shown in Figure 2.5). Even though unemployment decreased, labor market conditions deteriorated and real wages decreased, particularly in the informal labor market (IMF, 2024, p. 7). As shown in Figure 2.4, the percentage of poverty strongly increased.

After a period of instability, Javier Milei took office on December 10, 2023. In his first address, he outlined the economic issues facing Argentina, such as fiscal and trade deficits, risk of hyperinflation, rising debt, significant poverty, and stagnant growth (LA NACION,

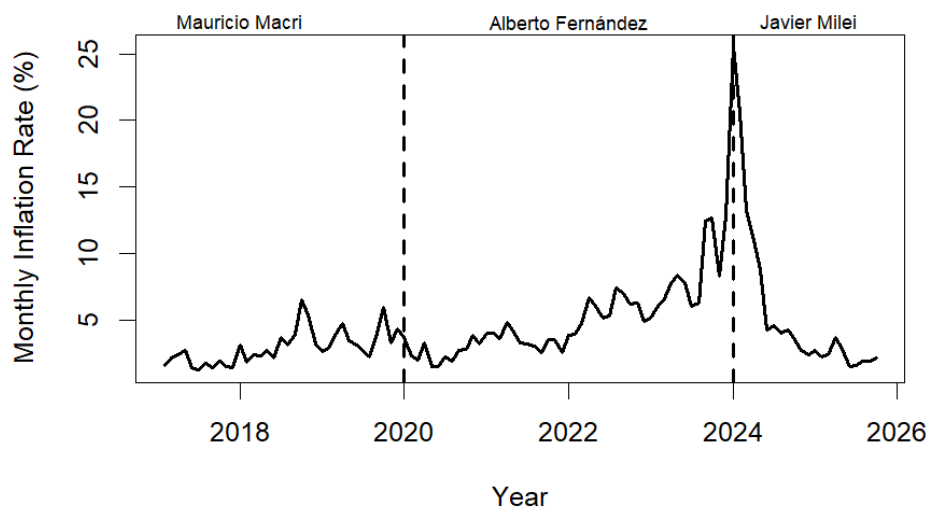


Figure 2.5: Inflation in Argentina (CPI) *Source:* INDEC (2025b)

2023, 5:11–15:00). Milei stated, “No hay plata” (“There is no money”) (LA NACION, 2023, 15:19) and emphasized the need for fiscal and monetary reforms to change the situation. He recognized that these reforms would come with challenges but insisted they were crucial for eventual recovery and long term growth (LA NACION, 2023, 15:50–17:30). The results of this package of policies were a reduction in monthly inflation rate, fiscal surplus, trade surplus, decrease in risk and increase in net international reserves (IMF, 2024, pp. 6–12). Nonetheless, this came with a cost: for example an initial decrease of real supermarket sales by 10% from November 2023 to March 2024, while at the same time real wages decreased by 17% (IMF, 2024, p. 7). In addition to these economic results, the adjustment programs caused social and political opposition. These happened through nationwide protests and strikes (Misculin, 2025; Phillips & Iglesia, 2024; Soria & Macarian, 2025). Nevertheless, on the 11th of April 2025, the country was able to transition from a fixed exchange rate system to a floating band regime, giving more flexibility to the exchange rate (IMF, 2025).

Chapter 3 Model Specification

"All models are wrong but some are useful"

— Box ([1979](#), p.2)

To answer the research question raised by this thesis, it is essential to create an appropriate model that allows the correct analysis of this topic. This chapter aims to first explain how VAR models work, then apply it to this specific case. Finally, it gives a brief look at how other studies have applied this model to similar questions.

Since Sims ([1980](#)) strongly advocated the use of multivariate equations, the field of macroeconomics started to use VAR extensively. According to him, the traditional methods used to connect the empirical reality and the theoretical models, known as the identification process, were often inappropriate, resulting in questionable claims. To solve this issue, he proposed the use of VAR models (Sims, [1980](#), p. 1). Thereafter, VAR models have been widely used in various fields of macroeconomics, particularly in the analysis of fiscal and monetary policy shocks.

3.1 Vector Autoregressive Models

The following subchapter is based on Kilian and Lütkepohl ([2017](#), pp. 23–25) to support the explanation of the VAR models.

Assume that the relationship between a vector of variables $y_t = (y_{1t}, \dots, y_{Kt})'$ and that the Data Generating Process (DGP) can be represented by two parts: by a deterministic

part (μ_t) and a purely stochastic one (x_t) such that

$$y_t = \mu_t + x_t \tag{3.1}$$

The deterministic component, μ_t , can take various forms, such as a constant term, polynomial components, seasonal terms or other dummy variables.

In contrast, the stochastic part, x_t , can be represented by a linear VAR

$$x_t = A_1 x_{t-1} + \dots + A_p x_{t-p} + u_t \tag{3.2}$$

where $A_i, i = 1, \dots, p$ are $K \times K$ coefficient matrices and $u_t = (u_{1t}, \dots, u_{kt})$ represents a K -dimensional white noise term with zero mean and covariance matrix $\mathbb{E}(u_t u_t') = \Sigma_u$.

Although VAR models usually require stationary data, Kilian and Lütkepohl (2017, pp. 106–108) note that estimation can be performed also in levels, which avoids possible unit root and cointegration restriction issues.

3.2 The Model

This chapter outlines the econometric model employed in the analysis. The model is a SVAR model, an extension of the VAR framework, which allows the construction of Impulse Response Functions (IRF). These are used to identify how shocks affect the variables in the model. The identification strategy is discussed further in Section 3.2.2.

3.2.1 Vector of Variables

The model employs a vector, x_t , of four endogenous variables. These variables represent key aspects of the Argentine economy.

$$X_t = \begin{bmatrix} T_t \\ G_t \\ Y_t \\ R_t \end{bmatrix} = \begin{bmatrix} \text{Tax Revenues} \\ \text{Government Spending} \\ \text{Output} \\ \text{Sovereign Risk} \end{bmatrix} \quad (3.3)$$

As shown in equation 3.3, X_t is a four dimension vector consisting of the logarithms of real per capita Tax Revenues (T_t), Government Spending (G_t), Output (Y_t) and Sovereign Risk (R_t), which allows the model to capture the dynamic effect between these variables.

3.2.2 Identification Strategy

The purpose of this subsection is to show the necessity of an identification strategy and to decide which one is more suitable. To do this, the representation in Kilian and Lütkepohl (2017, pp. 216–219) is presented.

Let's assume that the DGP of the SVAR model can be well approximated by the following form

$$B_0 x_t = B_1 x_{t-1} + \dots + B_p x_{t-p} + w_t, \quad (3.4)$$

where the deterministic part has been dropped for notational convenience. Notice that B_0 is the contemporaneous interaction matrix between the variables. On the other hand, w_t represents a zero-mean uncorrelated error term referred to as a structural shock. By multiplying both sides by B_0^{-1} we obtain

$$x_t = A_1 x_{t-1} + \dots + A_p x_{t-p} + u_t, \quad (3.5)$$

where $A_i = B_0^{-1} B_i$ and $u_t = B_0^{-1} w_t$. Note that this is the same equation as in 3.2

Before estimating the following model, it is needed to understand how to recover the structural form of the model. If B_0 is known, then it is possible to reconstruct $w_t = B_0 u_t$. From this, the variance of u_t can be used

$$\Sigma_u = \mathbb{E}(u_t u_t') = B_0^{-1} \mathbb{E}(w_t w_t') B_0^{-1'} = B_0^{-1} \Sigma_w B_0^{-1'} = B_0^{-1} B_0^{-1'}. \quad (3.6)$$

Since we assume that the structural shocks in w_t are uncorrelated and each of them is normalized to one, we can use $\Sigma_u = I_K$. From $\Sigma_u = B_0^{-1} B_0^{-1'}$, I can create a system of non-linear equations and find the values of B_0^{-1} . However, this is only possible if the number of parameters of B_0^{-1} is not larger than the number of independent equations. This requires imposing restrictions on B_0^{-1} .

Recall the definition used in equation 3.5 of $u_t = B_0^{-1} w_t$. This can equivalently be written as

$$B_0 u_t = C w_t \quad (3.7)$$

where $\Sigma_w = I_K$ such that $\Sigma_u = B_0^{-1} C C' (B_0^{-1})'$. This representation is very convenient, as it relaxes the assumption that either B_0 or C must be equal to the identity matrix.

I will now analyze two different methodologies to solve this problem and decide which one is more adapted.

Recursively Identified Models

To isolate the structural innovations w_t , a possible solution is the orthogonalization of the reduced-form errors u_t . This can be achieved using a lower triangular Cholesky decomposition of Σ_u . However, while mathematically applicable for every model, because we impose a strict causal chain of the nature of causal relationships from the data, it must also make sense economically (Kilian & Lütkepohl, 2017, pp. 219–220).

I present the matrix representation of equation 3.2.

$$\begin{bmatrix} u_t^{tax} \\ u_t^{gov} \\ u_t^{gdp} \\ u_t^{risk} \end{bmatrix} = \begin{bmatrix} b_{11} & 0 & 0 & 0 \\ b_{21} & b_{22} & 0 & 0 \\ b_{31} & b_{32} & b_{33} & 0 \\ b_{41} & b_{42} & b_{43} & b_{44} \end{bmatrix} \begin{bmatrix} w_t^{tax} \\ w_t^{gov} \\ w_t^{gdp} \\ w_t^{risk} \end{bmatrix} \quad (3.8)$$

In this case, since only B_0 is unknown, we only need $\frac{k(k-1)}{2}$. With four variables this is equal to 6. As shown in equation 3.8, this can be achieved through a lower-triangular Cholesky decomposition of Σ_u . To understand if this is a suitable approach, I will expand the equations and see if they make sense economically.

$$u_t^{tax} = b_{11}w_t^{tax}$$

The reduced form residual of T, u_t^{tax} , depends only on its structural shock w_t^{tax} . This implies that during the current period, tax revenues are not immediately affected by other variables. Unfortunately, this assumption can be considered unrealistic, since automatic stabilizers imply that taxes respond contemporaneously to output movements.

$$u_t^{gov} = b_{21}w_t^{tax} + b_{22}w_t^{gov}$$

The reduced form residual of G, u_t^{gov} , depends on the contemporaneous structural shocks to w_t^{tax} and w_t^{gov} . This implies that fiscal authorities can respond contemporaneously to unexpected movements in Tax revenues, but not to output and risk.

$$u_t^{gdp} = b_{31}w_t^{tax} + b_{32}w_t^{gov} + b_{33}w_t^{gdp}$$

The reduced form residual of output, w_t^{gdp} , depends on the contemporaneous structural shocks of w_t^{tax} , w_t^{gov} and w_t^{gdp} . Economically, this means that output responds contemporaneously to shocks in taxes, government spending and its own structural shock. This, however, is incomplete. The risk variable could also have a contemporaneous effect on

the current output.

$$u_t^{risk} = b_{41}w_t^{tax} + b_{42}w_t^{gov} + b_{43}w_t^{gdp} + b_{44}w_t^{risk}$$

The reduced form of sovereign risk, w_t^{risk} , depends on the contemporaneous shocks of w_t^{tax} , w_t^{gov} , w_t^{gdp} and w_t^{risk} . This means that risk responds contemporaneously to shocks in taxes, government spending, output and on its own shock. This is plausible since sovereign risk can take each of these shocks as signals about the country's economy.

As demonstrated, the purely recursive identification strategy would be problematic since tax revenues and government spending, due to automatic stabilizers, do respond contemporaneously to shocks in output. On the other hand, output also does respond to taxes and government spending. Additionally, assuming that risk does not have a contemporaneous effect on output, would be a critical flaw of the model. Altering the variable order would only require similar or even weaker assumptions. Consequently, this strategy is not well suited for this analysis.

The Blanchard and Perotti (2002) identification strategy

The following matrix equation, adapted to answer the research question of this thesis, utilizes the identification strategy first developed by Blanchard and Perotti (2002).

$$\begin{bmatrix} 1 & 0 & b_{13} & 0 \\ 0 & 1 & b_{23} & 0 \\ b_{31} & b_{32} & 1 & b_{33} \\ b_{41} & b_{42} & b_{43} & 1 \end{bmatrix} \begin{pmatrix} u_t^{tax} \\ u_t^{gov} \\ u_t^{gdp} \\ u_t^{risk} \end{pmatrix} = \begin{bmatrix} 1 & c_{12} & 0 & 0 \\ c_{21} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} w_t^{tax} \\ w_t^{gov} \\ w_t^{gdp} \\ w_t^{risk} \end{pmatrix} \quad (3.9)$$

As before, the economic logic underlying the variables is analyzed

$$u_t^{tax} = w_t^{tax} + c_{12}w_t^{gov} - b_{13}u_t^{gdp}$$

This equation offers a formulation that is more economically consistent than the recursive

model. Now the reduced form residual u_t^{tax} not only depends on its structural shock w_t^{tax} , but also on $c_{12}w_t^{gov}$ and $b_{13}u_t^{gdp}$. c_{12} represents the coordination between spending and tax policies together. Specifically, when government spending changes, tax policy may be adjusted in the same period as part of a coordinated fiscal policy. On the other hand, b_{13} captures the automatic response of taxes to movements in output. For example, if output increases, tax revenue will also automatically increase. Due to restriction requirements, this is measured exogenously (see Chapter 3.3).

$$u_t^{gov} = c_{21}w_t^{tax} + w_t^{gov} - b_{23}u_t^{gdp}$$

The reduced form residual u_t^{gov} depends on its structural shock w_t^{gov} , on $c_{21}w_t^{tax}$ and $b_{23}u_t^{gdp}$. c_{21} , similarly to c_{12} , represents the coordination of tax and spending policies. However, to estimate the model, either c_{12} or c_{21} have to be set equal to zero. In turn, b_{23} represents the immediate response of government spending to changes in output. Blanchard and Perotti (2002) assume that governments respond with delay to output fluctuations and consequently it is equal to zero. In addition, automatic stabilizers in countries like Argentina are usually less effective as they have limited funding, design flaws and high rates of informality (Folgar & Posleman, 2024, p. 9). These two factors, are plausible arguments for this restriction. Nevertheless this assumption is verified empirically in Chapter 3.3.

$$u_t^{gdp} = w_t^{gdp} - b_{31}u_t^{tax} - b_{32}u_t^{gov} - b_{33}u_t^{risk}$$

The reduced form residual u_t^{gdp} depends on the structural output shock w_t^{gdp} and on the contemporaneous effects of taxes b_{31} , government spending b_{32} , and risk b_{33} . Therefore, output reacts immediately to fiscal and risk shocks. Risk shocks are relevant as they raise borrowing costs and indicate uncertainty in the country.

$$u_t^{risk} = w_t^{risk} - b_{41}u_t^{tax} - b_{42}u_t^{gov} - b_{43}u_t^{gdp}$$

The reduced form residual u_t^{risk} depends on the structural risk shocks w_t^{risk} , and on the contemporaneous effects of taxes b_{41} , on government spending b_{42} and on output b_{43} . Given Argentina's long history of fiscal budget imbalances, government spending and taxes could be important determinants of the sovereign risk.

3.2.3 Structural Vector Autoregressive Models in the Literature

The methodology first applied by Blanchard and Perotti (2002) has been used in a number of studies. Their analysis, focusing on the United States from 1947 to 1997, show that expansionary fiscal shocks do increase output. In particular, following an increase in government spending, private consumption reacts positively while private investments negatively. Specifically, the effect reaches a peak after almost four years with a multiplier equal to 1.29 (Blanchard & Perotti, 2002, pp. 1345–1347). On the other hand, tax shocks have a negative effect on output, generally reaching the peak effect around five quarters after the shock and recovering afterwards. The measured multiplier is equal to -0.78 after five periods (Blanchard & Perotti, 2002, pp. 1342–1345).

Auerbach and Gorodnichenko (2012) expand this methodology by utilizing regime-switching SVAR. Specifically, they estimate the fiscal multipliers depending on the business cycle, finding significant differences depending on the state of the economy. Government spending, as theory suggests, is much more effective in recessions compared to expansions. The multiplier in recessions is around 2.4, while in expansions it is approximately 0.5. However, as the authors state, these measures have to be taken carefully as they are the result of extreme cases (e.g. extremely bad recession) (Auerbach & Gorodnichenko, 2012, pp. 9–10).

Also Perotti (2004) extended Blanchard and Perotti (2002)'s study by examining five OECD countries (Australia, Canada, Germany, United Kingdom and USA). In addition to tax revenue, government spending and output they also used the GDP deflator inflation rate and the 10 year nominal interest rate. Interestingly, the effects of fiscal policy are found not to be too strong. For example, positive government spending multipliers larger than one after the 1980s are only found in the US. Specifically, government spending

shocks and tax cuts become weaker, while even showing negative effects on GDP, private consumption and private investment. In addition to this, he also did not find that tax cuts react quicker or have bigger effects than government spending policies.

Another extension of Blanchard and Perotti (2002) is provided by Tenhofen, Wolff and Heppke-Falk (2010) which studied the effect of different government subgroups (e.g. compensation of public employees) in Germany. Generally, fiscal policies do not have long run effects apart for an increase in public investment. They found that direct government expenditure increases output and private consumption with however a multiplier smaller than one and declining quickly (the effects reach zero after three years). Government operating expenditure is found to have a sizable positive effect on output, specifically because of capital formation. Compensations of public employees does not stimulate the economy. On the other hand, government net revenue shocks do not affect output significantly. Direct taxes increase significantly output, whereas indirect taxes do not have a significant effect.

Alternatively Alesina et al. (2019) in extend the narrative approach by Romer and Romer (2010) to identify exogenous fiscal consolidations. Their methodology only considers fiscal adjustments that were adopted exclusively to reduce the budget deficit and improve fiscal sustainability, rather than cooling down the economy. In addition, they recognize that fiscal policy measures are generally implemented through multiyear programs, rather than a one time shock. They also distinguish between unexpected and announced policies, which are likely to affect the expectations prior to implementation. Their results show that, on average, programs composed mostly by spending cuts deviated GDP by only 0.5 % from its usual growth rate only for roughly two years. In certain cases, they even found such cuts to be expansionary. On the other hand, tax based increases were found to be much more harmful: a 1% of deficit reduction lead to an output loss between 2 and 3% lasting for several years. Moreover, they found that expenditure based cuts do actually fulfill their premise and decrease the debt to GDP ratio, whereas tax based plans tend to increase the ratio.

3.3 The Data

In this section, I briefly discuss the data and variables utilized in this study. The code used to analyze and create the necessary plots of this thesis can be found in the Appendix. I used quarterly data ranging from 2004:1–2025:1. The output data and GDP deflator were taken from INDEC ([2025a](#)), the Argentine national statistical office. The deflator uses Q1 2004 as base period. The fiscal data stem from the Argentine Economic Ministry (Ministerio de Economía, [n.d.](#)) and its variables are cash data. Note that, due to quarterly data unavailability, the data refers to the central government and not the general government. As in Blanchard and Perotti ([2002](#)), all these variables have been transformed to real per capita terms. These variables have also been seasonally adjusted by using the X-13ARIMA-Seasonal adjustment strategy.

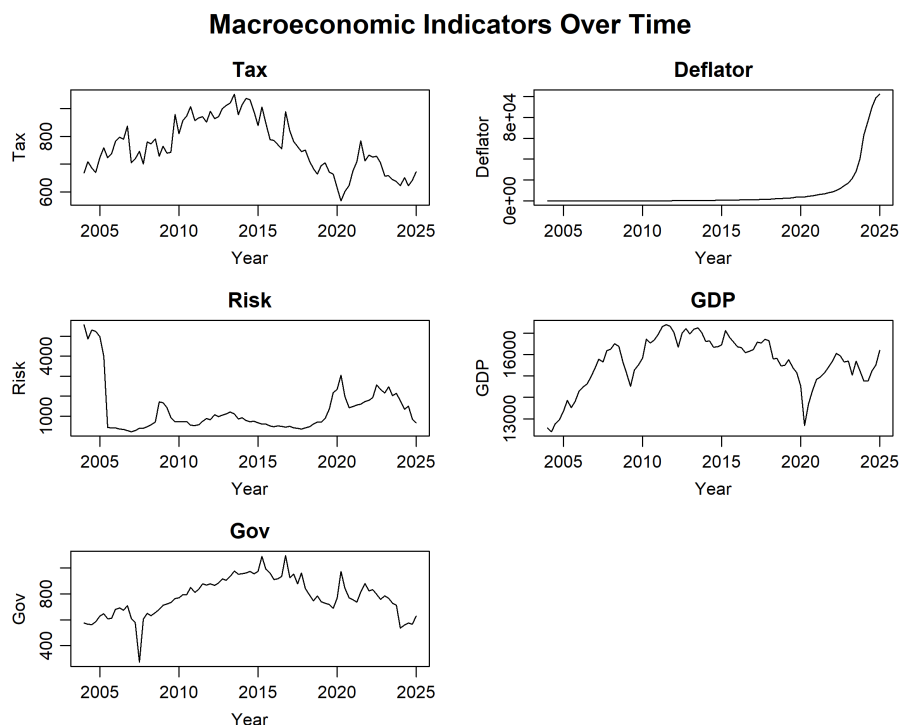


Figure 3.1: Macroeconomic Indicators over Time

The sovereign risk variable is represented by the "Riesgo País", which is an indicator that measures the riskiness of investing in Argentine government bonds compared to U.S. Treasury Bonds. It basically reflects the trust that international investors have in the

ability of the Argentine Government to repay its debt. This measure was first developed by J.P. Morgan and has become one of the most followed economic indicators in Argentina. It is calculated as the difference between Argentine sovereign bonds valued in dollars with the U.S. Treasury Bonds. The data stems from *Ámbito Financiero* ([n.d.](#)). This dataset originally has a daily frequency. To match the quarterly data used in the model, the average daily value for the quarter has been calculated to better represent the quarterly dynamics. In addition, all variables have been transformed into logarithms.

In Figure 3.1 the absolute real values of each macroeconomic variable are shown. Tax revenues show a continuous increase until around 2016, after which a decline can be observed reaching its lowest level during the pandemic. The deflator shows one of the most persistent issues in Argentina's economic history. Although in the early years of the series it seems relatively flat, this is only due to the extremely high levels of inflation experienced in the latter years: from 2004 to 2019 the average quarterly inflation was around 6%, while from 2020 to 2025 it averaged about 17%. The sovereign risk indicator shows extremely high values caused by the 2001 crisis in Argentina, seeing a substantial decrease in the subsequent years. In the latter years, however, it is still possible to see significant fluctuations. Real GDP per capita grows consistently for the initial years and is then followed by a large contraction caused by the financial crisis of 2008. Although partially recovering, it never reaches its peak again. From this point onward, real GDP per capita shows a mild decline, showing a big dip with the later pandemic. Government spending, in contrast, initially rises steadily and then remains at a fairly high level of government spending before declining in the mid-2010s, where it starts declining. The recent drop in expenditure is associated with the fiscal policy implemented by Argentina's current president Javier Milei.

The lag selection for the SVAR model has been based on information criterion for lag-order selection. Specifically, the Akaike Information Criteria ($AIC(n)$), the Hannan-Quinn Information Criterion ($HQ(n)$), the Schwarz Information Criterion ($SC(n)$) and the Final Prediction Error ($FPE(n)$) were used.

	1	2	3	4	5
AIC(n)	1.774230e+01	1.767150e+01	1.796440e+01	1.81133e+01	1.817390e+01
HQ(n)	1.798260e+01	1.810410e+01	1.858930e+01	1.89303e+01	1.918330e+01
SC(n)	1.834210e+01	1.875130e+01	1.952410e+01	2.01528e+01	2.069340e+01
FPE(n)	5.077629e+07	4.746571e+07	6.413675e+07	7.55715e+07	8.233178e+07

Table 3.1: VAR Lag Selection Criteria

As can be observed in Table 3.1, AIC and FPE suggest a lag equal to one, whereas HQ and SC suggest it to be equal to two. The main results will be presented with a lag equal to one, however, as a robustness check, also a lag equal to two will be used.

In addition, the baseline model assumes c_{12} is equal to zero. Later, as a robustness check, this restriction is set on c_{21} and the resulting changes in the estimates are observed and compared.

As described in Subchapter 3.2.2, I need to estimate b_{13} to identify immediate elasticity to changes to output. To this end, the following equation is estimated:

$$\Delta \ln(T_t) = \alpha_0 + \alpha_1 \ln(Y_t) + \epsilon_t \quad (3.10)$$

where T_t represents the Total Taxes at time t , Y_t the income at time t .

Similarly, to verify the assumption that governments do not act contemporaneously to shocks in output, the following equation is estimated:

$$\Delta \ln(G_t) = \beta_0 + \beta_1 \ln(Y_t) + \epsilon_t \quad (3.11)$$

where G_t is the total government spending at time t .

The precise results of the following regressions are presented in Appendix A. Total Taxes elasticity on output is statistical significant on a significance level equal to 1%. On the other hand, government spending elasticity does not result in any significant result. Consequently, the assumption of $b_{23} = 0$ has empirical support.

Chapter 4 Results

"It's not the figures themselves; it's
what you do with them that matters."

— Kennedy (2003, p. 2)

In this section, I will analyze the model's results using Impulse Response Functions and tables that represent how various shocks affect the different variables. Based on this, some considerations regarding the Argentine economy will be discussed.

4.1 The Contemporaneous Coefficients

Table 4.1 presents the contemporaneous coefficients estimated. Note that, by construction the variables b_{ij} enter the equations in Chapter 3.2.2 with a negative sign. Thus, it is important to remember that an estimated variable with a negative sign actually implies a positive effect, while a negative value implies a positive effect. b_{13} , the elasticity of tax revenue to output, and b_{23} , the immediate response of government spending to output, were exogenously estimated as shown in Chapter 3.3. b_{31} represents the effect of an increase in taxes on output. As an isolated result, it shows that the model behaves as theory predicts, as an increase in taxes has a negative effect on output. Similarly, b_{32} , the contemporaneous effect government spending on GDP is positive, which is also consistent with theory. b_{34} , the contemporaneous effect of risk with GDP, shows that an increase in sovereign risk has a negative effect on GDP. b_{41} , on the other hand, showcases the contemporaneous effect of taxes on risk. Specifically, the estimated value shows that an increase in taxes reduces risk. Interestingly, this is not the case for government spend-

4.2. The Effects of Government Spending Shocks

ing, which increases risk. Similarly, this is true for GDP. While this is counterintuitive, there are different possible explanations which will be further explored. c_{21} on the other hand represents the policy coordination between tax increases and government spending. The contemporaneous effect shows that government spending also increases when taxes increase.

	b_{13}	b_{23}	b_{31}	b_{32}	b_{34}	b_{41}	b_{42}	b_{43}	c_{21}
coeff.	-0.6076	0	3.2562	-0.2266	3.2577	3.6818	-0.1739	-38.5803	0.43800

Table 4.1: Estimated Contemporaneous Coefficients

4.2 The Effects of Government Spending Shocks

Table 4.2 shows the average response of the variables to a government spending shock over time.

	1 qrt	3 qrts	6 qrts	9 qrts	12 qrts
GDP	-0.0235	-0.0017	0.0223	0.0267	0.0229
Tax	0.0868	0.0828	0.0507	0.0348	0.0263
Risk	0.0278	-0.2454	-0.3587	-0.2840	-0.1818

*Note: The * indicates whether the response is significant (10%).*

Table 4.2: Response to Government Spending

In Figure 4.1, the Impulse Response Functions to a shock in Government Spending are shown. Note that on the x-axis the quarters are shown, while on the y-axis the logged values of the respective variables are shown. This means that the variations are always relative to their baseline. The x-axis starts with period one, which represents the contemporaneous effect of the shock. The solid black line represents the Impulse Response Function, which represents the average response. Note that if the dotted lines, which represent the confidence intervals, contain the zero x-line, the effects are insignificant, meaning that it is not possible to show a causal effect. The impulse functions respond to a one-standard-deviation structural shock.

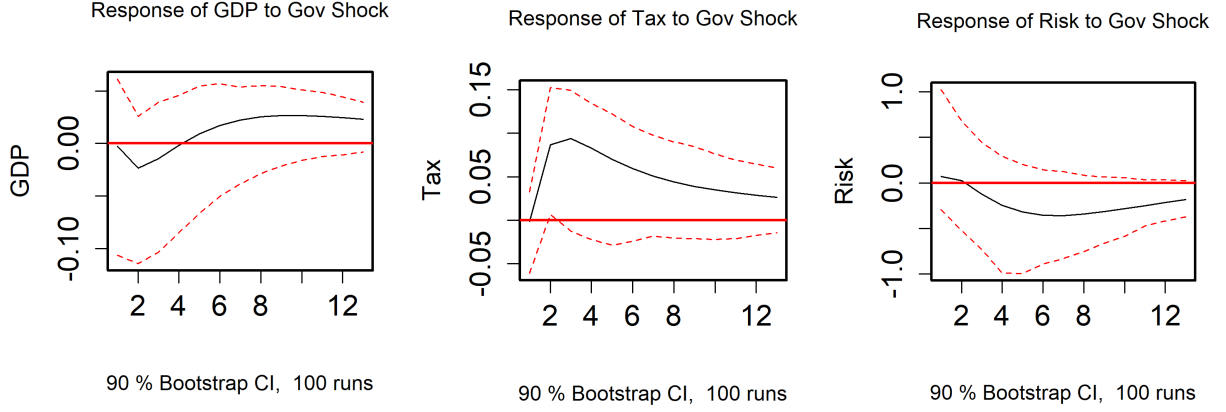
**Figure 4.1:** Effects of Government Spending Shocks

Figure 4.1 shows some interesting findings. Contrary to standard economic theory, government spending shocks have no significant effect on output. Additionally, also the mean response shows an interesting behavior. Initially it decreases while becoming slightly positive in the later quarters. Similarly, tax revenues show no causal relationship with government spending shocks. Since government spending does not increase output, it is a logical consequence that also tax revenue does not increase significantly. Risk also shows an insignificant response.

4.3 The Effect of Tax Shocks

	1 qrt	3 qrts	6 qrts	9 qrts	12 qrts
GDP	0.2032*	0.2525*	0.2246*	0.1755*	0.1291*
Gov	0.8900*	0.7676*	0.4942	0.3513	0.2672
Risk	-1.8639*	-2.0022	-1.4826	-0.9426	-0.5576

*Note: The * indicates whether the response is significant (10%).*

Table 4.3: Response to Tax Shock

Figure 4.2 and Table 4.3 showcase further interesting results. Contrary to standard economic theory, a shock in tax revenue is followed by significant economic growth, peaking at 0.2525 three quarters later. This is significant for all three years estimated. With a shock in taxes, also government spending increases reaching its peak at 0.89 one quarter after the implementation of the tax hike. This effect is significant until the fifth quarter (included). Interestingly, also risk decreases with a tax shock. This is significant from

4.4. The Effect of GDP Shocks

one to two quarters after the policy implementation.

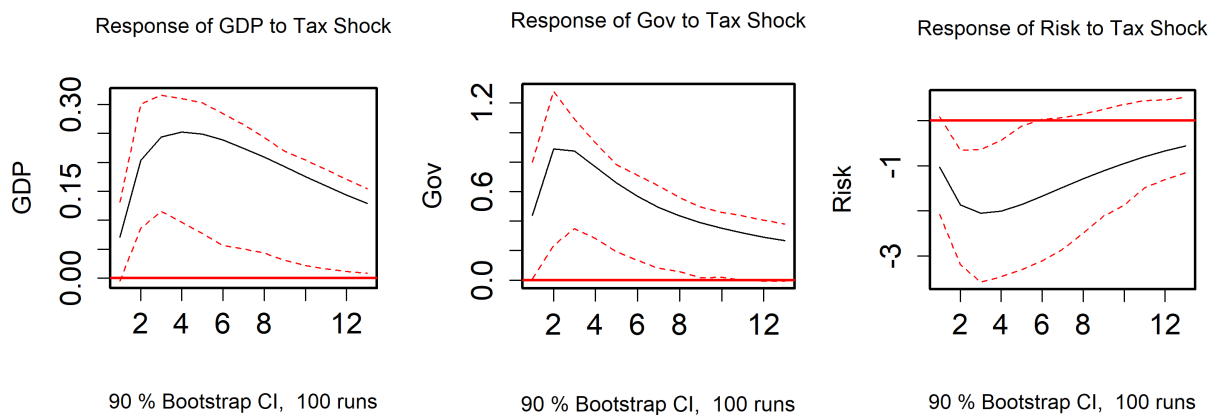


Figure 4.2: Effects of Tax Shocks

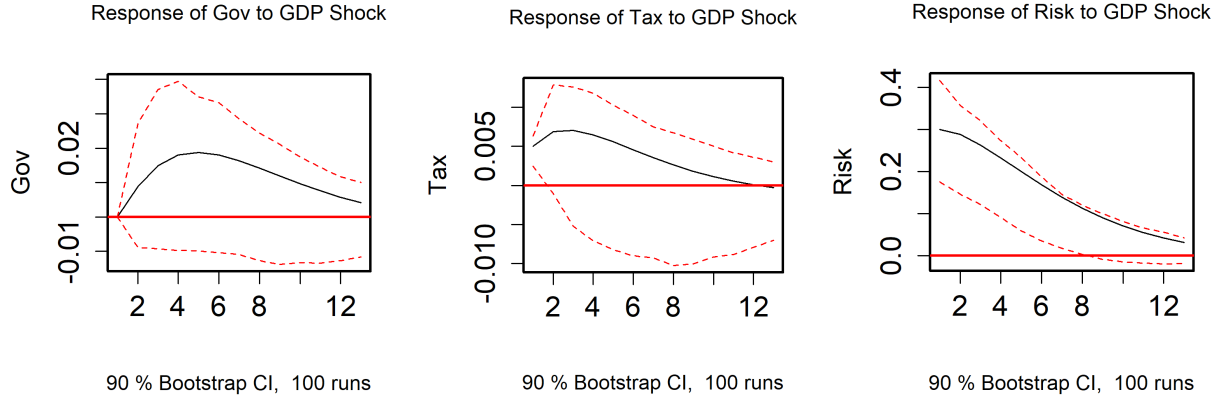
4.4 The Effect of GDP Shocks

	1 qrt	3 qrts	6 qrts	9 qrts	12 qrts
Gov	0.0089	0.0180	0.0164	0.0097	0.0042
Tax	0.0069	0.0065	0.0035	0.0011	-0.0003
Risk	0.2877*	0.2324*	0.1392*	0.0709	0.0312

*Note: The * indicates whether the response is significant (10%).*

Table 4.4: Response to GDP shock

The GDP shock also showcases interesting results. While government spending and tax revenue react insignificantly to such a shock, the same is not true for risk. Surprisingly, risk initially increases and then decreases over time. This is significant until two years after the policy implementation, peaking immediately when the shock happens. This pattern could be due to the effect of expectations on the economy. If the risk variable had already reacted in anticipation, then it is plausible that most of the adjustment had already happened before the shock occurred.

**Figure 4.3:** Effects of GDP Shocks

4.5 The Effects of Risk Shocks

	1 qrt	3 qrts	6 qrts	9 qrts	12 qrts
GDP	-0.0207*	-0.0124*	-0.0060	-0.0033	-0.0022
Gov	-0.0087	-0.0189	-0.0211	-0.0171	-0.0123
Tax	-0.0171*	-0.0161*	-0.0127*	-0.0091	-0.0063

*Note: The * indicates whether the response is significant (10%).*

Table 4.5: Response to Risk shock

Figure 4.4 and Table 4.5 showcase the results after a sovereign risk shock. An increase in risk causes an immediate significant decrease in output. This effect is significant until and including the fifth quarter, peaking immediately with a value of -0.0268. Tax revenue also reacts negatively and significantly until the seventh quarter peaking one quarter after the policy implementation with a value equal to -0.0171. Interestingly, government response to risk shock is not statistically significant. Nevertheless, the average behavior shows an interesting pattern: even if insignificant, the mean response is clearly not positive. This in part suggests that Argentine governments acknowledge that increasing spending would further increase the sovereign risk.

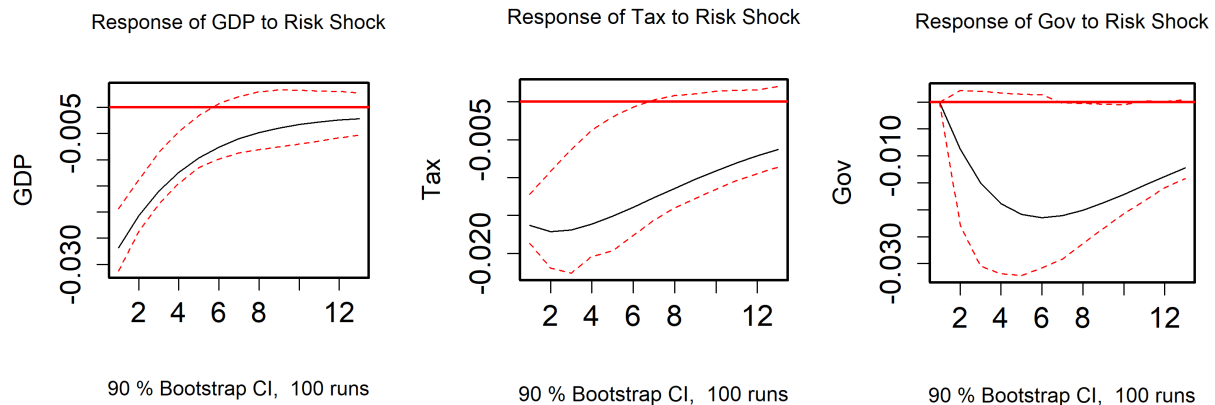


Figure 4.4: Effects of Risk Shocks

4.6 Robustness Checks

To assess the validity of the model, I conduct a series of robustness checks. The corresponding results are presented as IRFs and tables in the Appendices which replicate what was done in the previous subchapter. First of all, since the elasticity of tax revenue on output is externally estimated, alternative measures are taken to understand how sensitive the results are. Initially, I will adjust the elasticity by 0.25 positively and negatively from the estimated value, and then re-estimate the models. As an additional measure, I also use the estimated short-term elasticity measured in Machado and Zuloeta (2012, p. 22). Even though the period for which this elasticity is calculated does not cover the entire time of this research, it still could be useful as a comparison. This will also further show the effects that the short-term elasticity has on the overall results. The estimated value is equal to 1.21. Moreover, the models are re-estimated by changing the identification strategy by lifting the zero restriction on c_{12} and instead imposing it on c_{21} . Finally, as noted in Subchapter 3.3, the lag length may be changed to two. As a consequence, I will check if this changes the results in any significant way. The results are summarized in Table 4.6

4.7 Results Discussion

This section seeks to interpret how fiscal variables interact with risk and how they jointly influence output in Argentina based on the SVAR results shown earlier. To interpret these

	Gov Shock	Tax Shock	GDP Shock	Risk Shock
Elasticity +0.25	Comparable results	Comparable results	Comparable result	Comparable results. Government spending decreases with an increase in risk shock from quarters 2 to 8
Elasticity -0.25	Comparable Results	Comparable Result, but the effect on risk becomes insignificant	Comparable Results	Comparable Results
Elasticity = 1.21	Comparable Results	Similar results, but GDP effects are smaller, and Risk responses are longer significant (quarters 1–8).	Comparable Results	Comparable Results. Taxes react stronger and government spending decreases with an increase in risk shock (from quarter 2 to 10)
c restriction	Comparable Results	Comparable results, but Tax Shock is significant for a lower length	Comparable result	Comparabale Result
lag = 2	Comparable Result	Comparable Result but the effect on risk becomes insignificant	Comparable Results	Comparable Results

Table 4.6: Summary of Robustness Checks

results, Table 4.1 gives a good starting point, from which the discussion will be developed.

4.7.1 Insignificance of Government Spending

As shown in the previous sections, government spending shocks showcase insignificant effects on all the other variables. The contemporaneous coefficients give an interesting hint why this is true. As shown with the contemporaneous coefficient b_{33} , an increase in risk decreases output. While government spending does have an initial positive effect on output (as shown by b_{32}), b_{42} shows that it also has a negative effect on risk. This shows that there is a counterbalancing effect between these two factors indicating that Argentina has limited fiscal space, showing strong crowding-out effects given the distrust that investors have in the country. Additionally, the relatively small effect of b_{32} , could indicate that the government spending in Argentina is quite unproductive.

4.7.2 Positive Output Effects of Tax Increases

Probably the most surprising results of this study are obtained here. Overall, tax increases have a positive output effect. However, this effect is not immediate. The contemporaneous effect of taxes on output, b_{31} , shows a negative effect, suggesting that, as economic theory suggests, it does decrease output. However, this is offset by other factors. First of all, as already mentioned, a decrease in risk increases output (b_{32}). Second, b_{41} shows that tax increase actually does decrease risk. This suggests that credibility effects come into play: since Argentina has had a long history of credibility issues, increasing taxes signals greater fiscal sustainability, reducing macroeconomic uncertainty and government stability.

Chapter 5 Conclusion and Policy Implications

“One could argue that economics is a particularly underdeveloped science, that there is still much we do not know.”

— Mankiw ([2006](#))

This thesis has the goal of analyzing how fiscal variables interact with sovereign risk and how they influence economic growth in Argentina. Over the last decades, Argentina has been a country marked by chronic fiscal imbalances and recurring credibility issues. This made Argentina an ideal candidate for the analysis of the effect of the risk channel on fiscal variables and economic growth. By expanding the SVAR model proposed by Blanchard and Perotti ([2002](#)) with the Riesgo País as a variable of sovereign risk, this thesis provides new evidence on how fiscal and risk shocks propagate through the economy.

The results found in this thesis show that government spending has an insignificant short run effect on output, tax revenues, and sovereign risk. Contrary to traditional economic literature, which generally shows a positive multiplier effect, these results align more closely with the literature on emerging markets, where multipliers tend to be lower. In the Argentine context, marked by high inflation rates, limited fiscal policy space and high economic uncertainty, these factors limit the stabilization effects of government spending.

On the other hand, tax shocks seem to have a significant expansionary effect. An increase

in tax revenues increases GDP and in most specifications it also significantly decreases the sovereign risk. These results suggest that in countries where there is a lot of uncertainty about fiscal credibility, fiscal consolidation can improve expectations, reduce uncertainty and thus support economic activity.

Risk shocks also are found to have an important economic consequence on the economy. As the economy becomes more risky, also output and tax revenues decrease. Depending on the specification, this can also decrease government spending, showing that governments are aware of the importance of fiscal stability to lower Argentina's riskiness.

Nevertheless, this study has several limitations. First, the study includes only partially the effects of expectations in the model. While the risk variable is an average of daily values of the Riesgo País, which intrinsically includes the expectations of a country, the model does not account for it directly. In such unstable countries, much information can have an influence on the expectations of a country. For example, expectations can be influenced by the presidential elections or by the inflation path the country is following. These are all important factors that affect sovereign risk measures. Additionally, the effect of announced and unannounced policies is not taken into account. Second, multipliers are state dependent. For example, in a recession the multiplier is different from the one in an expansion. This could show even stronger effects compared to this study, or even reverse them. However, due to data limitations, this cannot be done since the non-linear SVAR would require more data points to estimate the model. A possible extension of this model could be including a threshold for the public debt variable. This inability to extend the model brings another possible issue: having only 84 observations may not be sufficient and can cause estimation problems. Third, while the role of the central government is fundamental in the Argentine economy, so are also the local governments. Not including this data could give an incomplete picture. Finally, omitting important variables such as the inflation rate or the exchange rate may distort the results.

These limitations suggest that further research is required. Expanding the model to address the mentioned limitations by, for example incorporating a state dependent model,

would strengthen the analysis. Moreover, applying the model to other countries with similar characteristics could give a valuable comparison to understand if Argentina is really a unique economy, or other countries do experience similar patterns.

Despite these limitations, the findings of this study have important policy implications. First, government spending does not always have a positive effect on output. For fiscally unstable countries such as Argentina, where an increase in government spending can create distrust in its fiscal stability, it may be beneficial to not increase spending. While this thesis did not study the composition of government spending, this may also suggest that focusing on productive government spending should be a priority. Second, policies that promote fiscal stability can be expansionary. For example, as this thesis shows, an increase in taxes can reduce risk and promote economic growth. Nevertheless, this result has to be taken cautiously. To increase taxes for an even bigger increase in government spending is probably not very beneficial as it does not promote fiscal soundness. This suggests that prioritizing medium to long term fiscal consolidation would be beneficial for the long term growth of Argentina.

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Appendix A - Elasticity models

Regression Results	
<i>Dependent variable:</i>	
	Tax
GDP	0.6076*** (0.2013)
Constant	-0.0017 (0.0060)
Observations	84
R ²	0.1000
Adjusted R ²	0.0890
Residual Std. Error	0.0545 (df = 82)
F Statistic	9.1062*** (df = 1; 82)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

(a) Tax elasticity model

Regression Results	
<i>Dependent variable:</i>	
	Gov
GDP	0.1791 (0.5242)
Constant	0.0005 (0.0156)
Observations	84
R ²	0.0014
Adjusted R ²	-0.0108
Residual Std. Error	0.1420 (df = 82)
F Statistic	0.1168 (df = 1; 82)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

(b) Government spending elasticity model

Figure A.1: Comparison of Elasticity Models

Appendix B - Higher Elasticity Specification

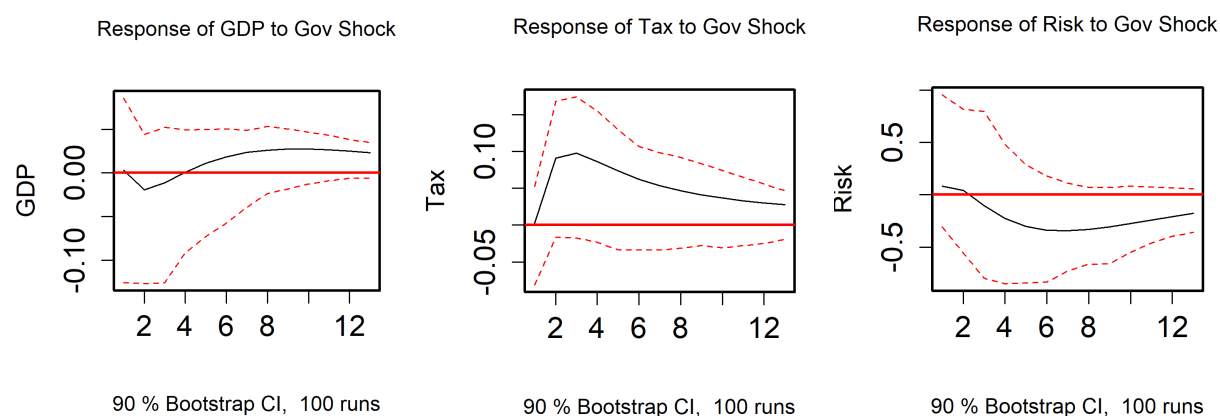


Figure B.2: Effects of Government Spending Shocks - Higher Elasticity

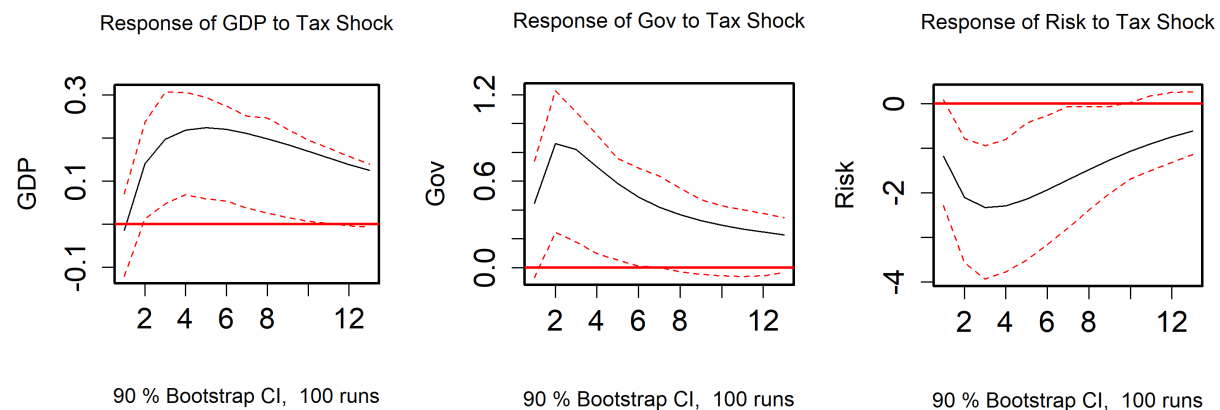


Figure B.3: Effects of Tax Shocks - Higher Elasticity

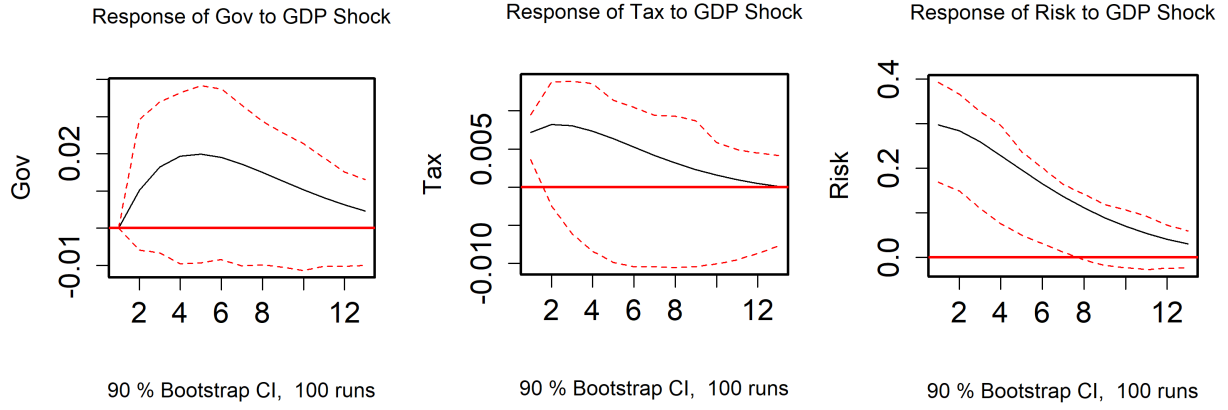


Figure B.4: Effects of GDP Shocks - Higher Elasticity

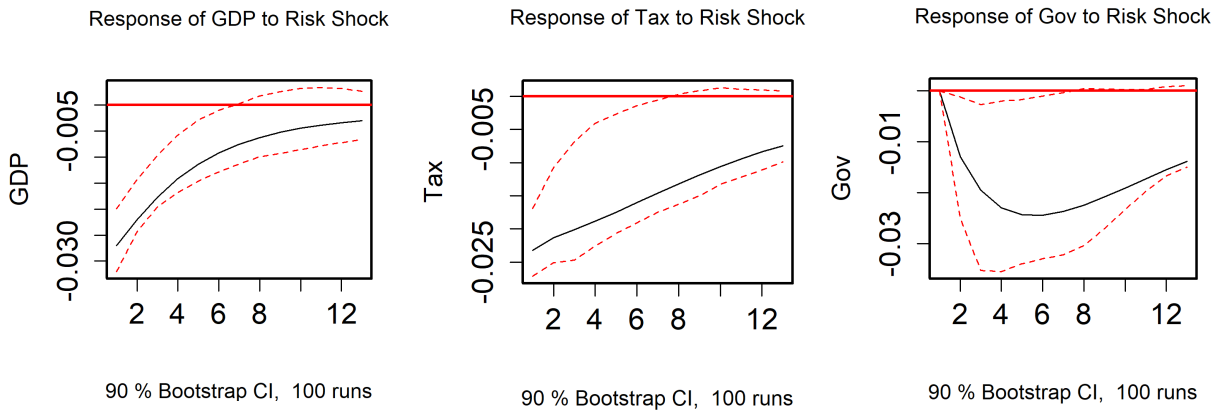


Figure B.5: Effects of Risk Shocks - Higher Elasticity

	b_{13}	b_{23}	b_{31}	b_{32}	b_{34}	b_{41}	b_{42}	b_{43}	c_{21}
coeff.	-0.8576	0	4.0113	-0.2755	3.243	0.6780	0.01680	-36.1832	0.4462

Table B.1: Estimated Contemporaneous Coefficients - Higher Elasticity

Appendix C - Lower Elasticity Specification

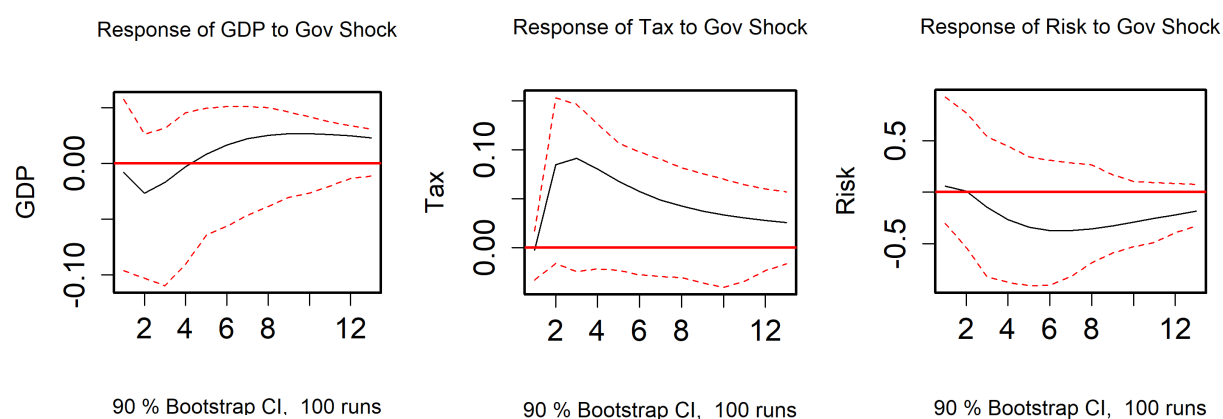


Figure C.6: Effects of Government Spending Shocks - Lower Elasticity

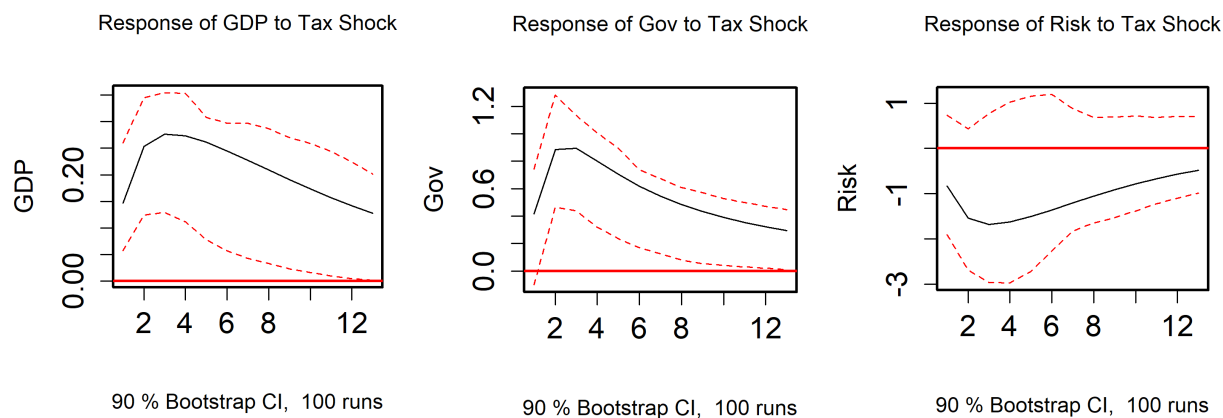


Figure C.7: Effects of Tax Shocks - Lower Elasticity

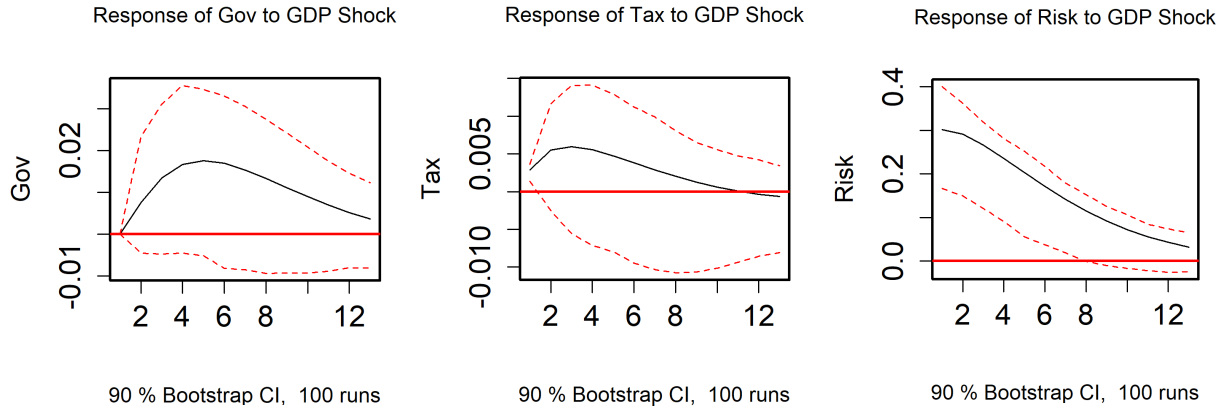


Figure C.8: Effects of GDP Shocks - Lower Elasticity

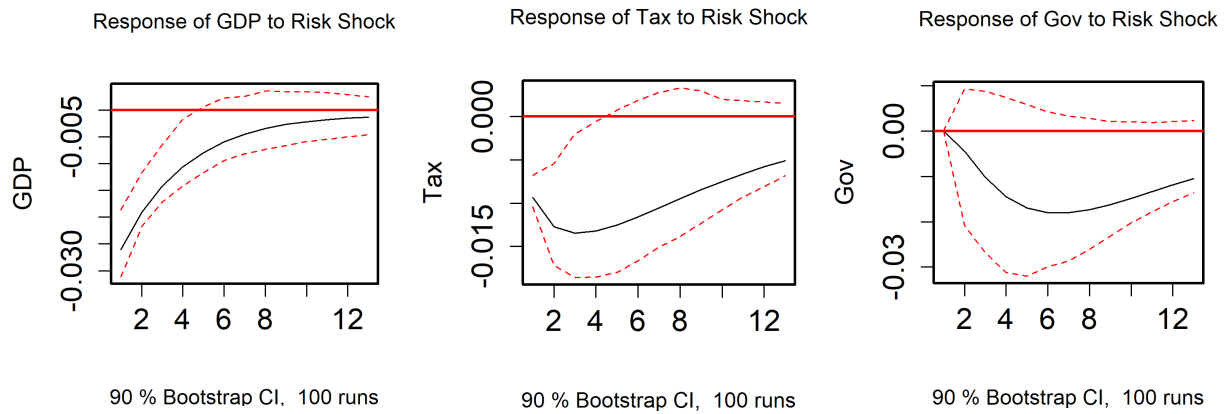


Figure C.9: Effects of Risk Shocks - Lower Elasticity

	b_{13}	b_{23}	b_{31}	b_{32}	b_{34}	b_{41}	b_{42}	b_{43}	c_{21}
coeff.	-0.3576	0	2.4971	-0.1773	3.2672	6.5088	-0.3548	-40.0777	0.4145

Table C.2: Estimated Contemporaneous Coefficients - Lower Elasticity

Appendix D - Alternative Elasticity Specification

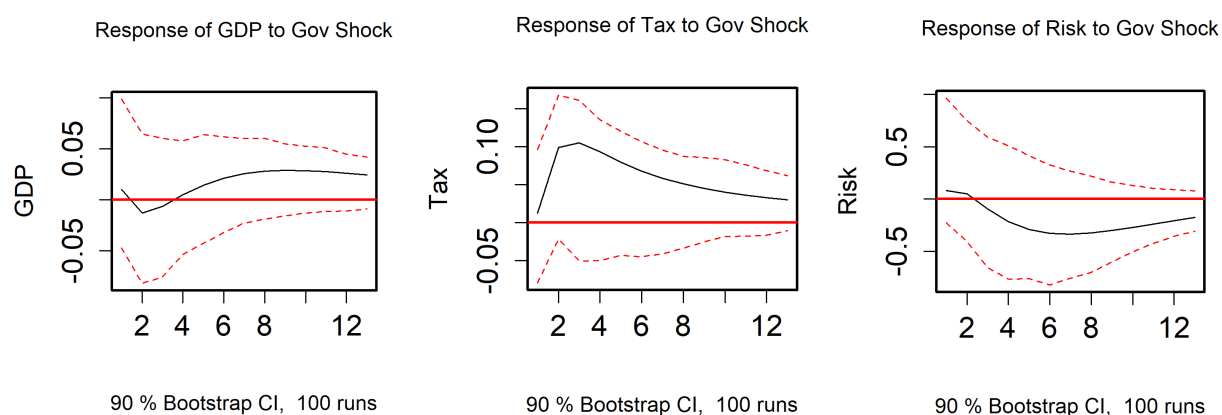


Figure D.10: Effects of Government Spending Shocks - Alternative Elasticity

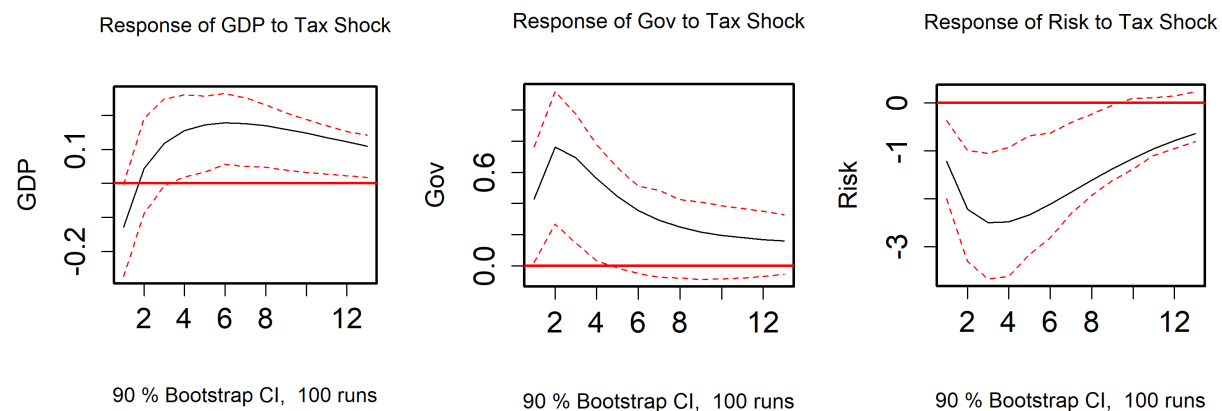


Figure D.11: Effects of Tax Shocks - Alternative Elasticity

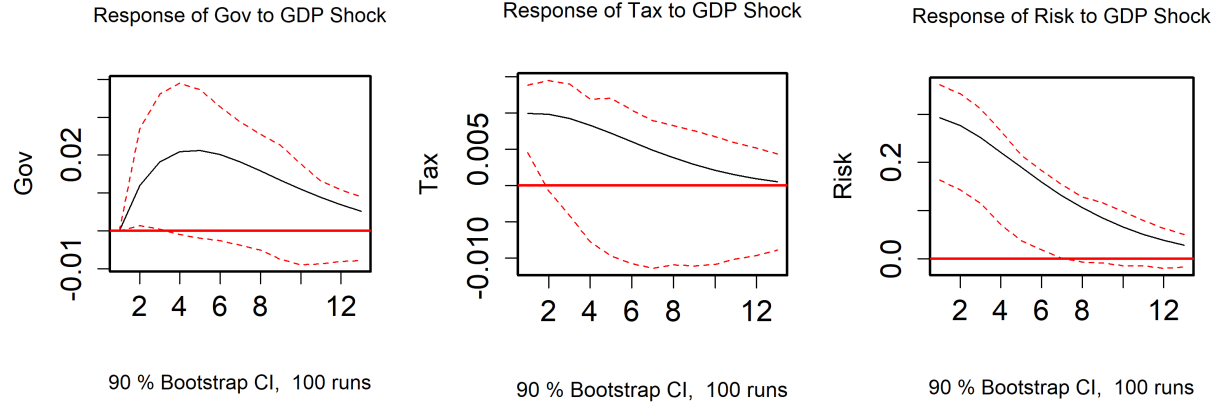


Figure D.12: Effects of GDP Shocks - Alternative Elasticity

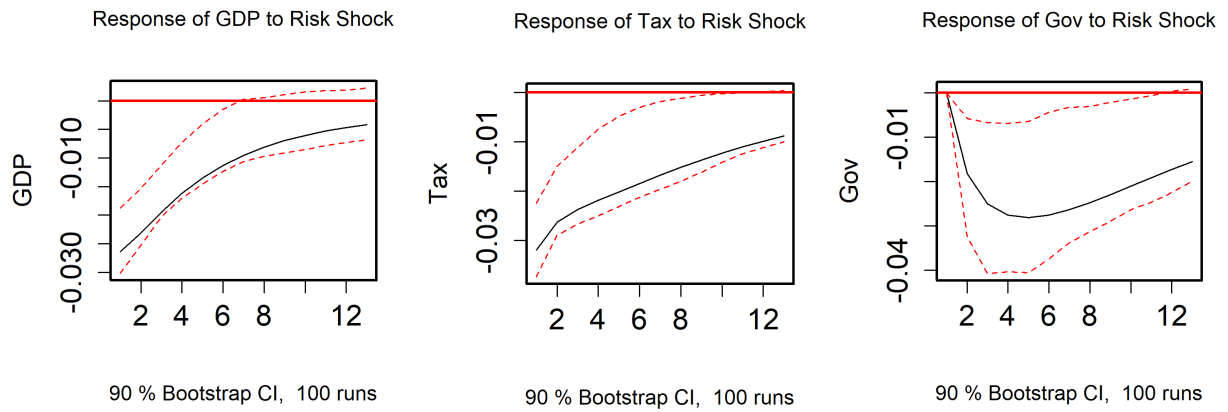


Figure D.13: Effects of Risk Shocks - Alternative Elasticity

	b_{13}	b_{23}	b_{31}	b_{32}	b_{34}	b_{41}	b_{42}	b_{43}	c_{21}
coeff.	-1.210	0	4.9798	-0.3383	3.2163	-3.4965	0.2796	-31.518	0.4284

Table D.3: Estimated Contemporaneous Coefficients - Alternative Elasticity

Appendix E - Alternative c Specification

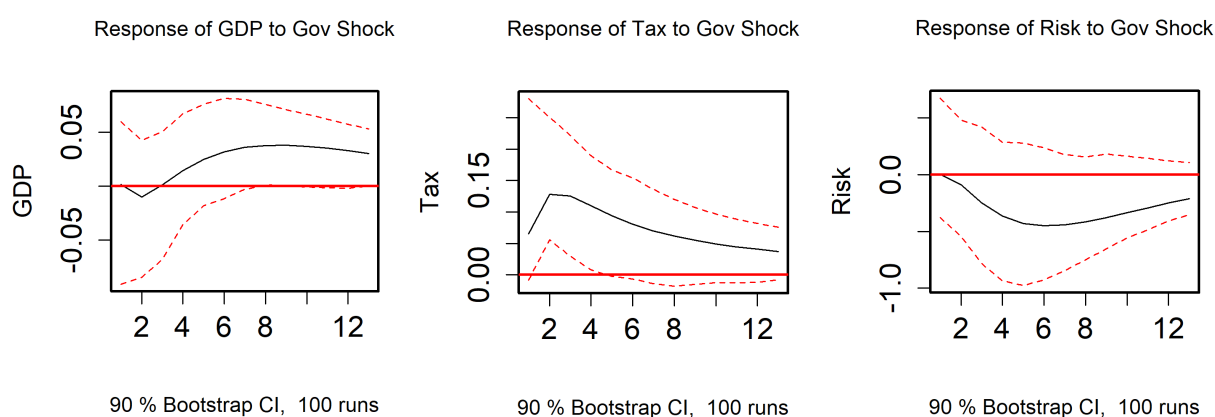


Figure E.14: Effects of Government Spending Shocks - Alternative c

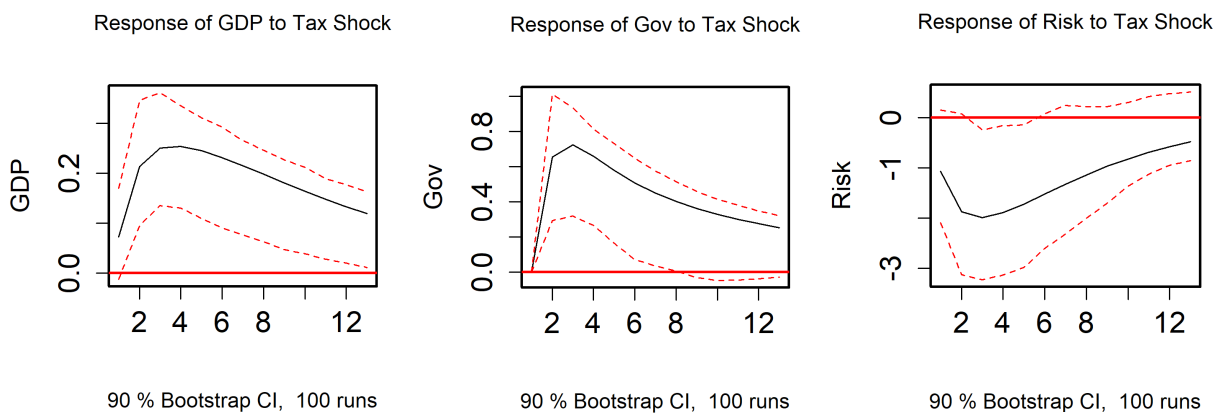


Figure E.15: Effects of Tax Shocks - Alternative c

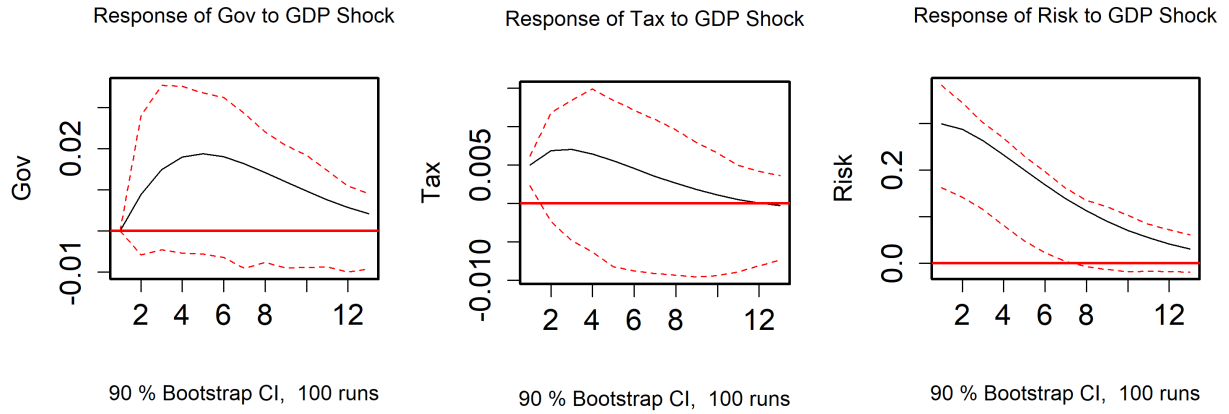


Figure E.16: Effects of GDP Shocks - Alternative c

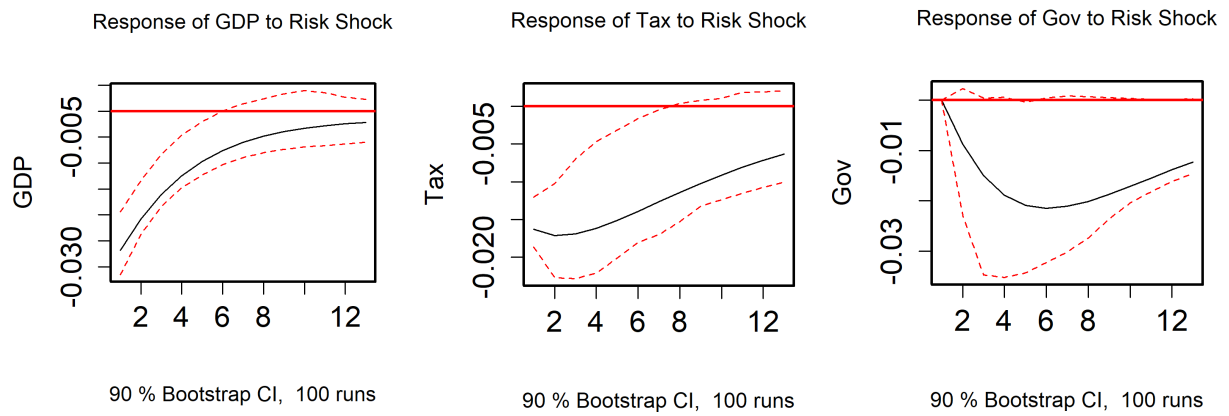


Figure E.17: Effects of Risk Shocks - Alternative c

	b_{13}	b_{23}	b_{31}	b_{32}	b_{34}	b_{41}	b_{42}	b_{43}	c_{12}
coeff.	-0.6076	0	3.2562	-0.2266	3.2577	3.6817	-0.1739	-38.5802	0.0639

Table E.4: Estimated Contemporaneous Coefficients - Alternative c

Appendix F - Extended Lag Specification

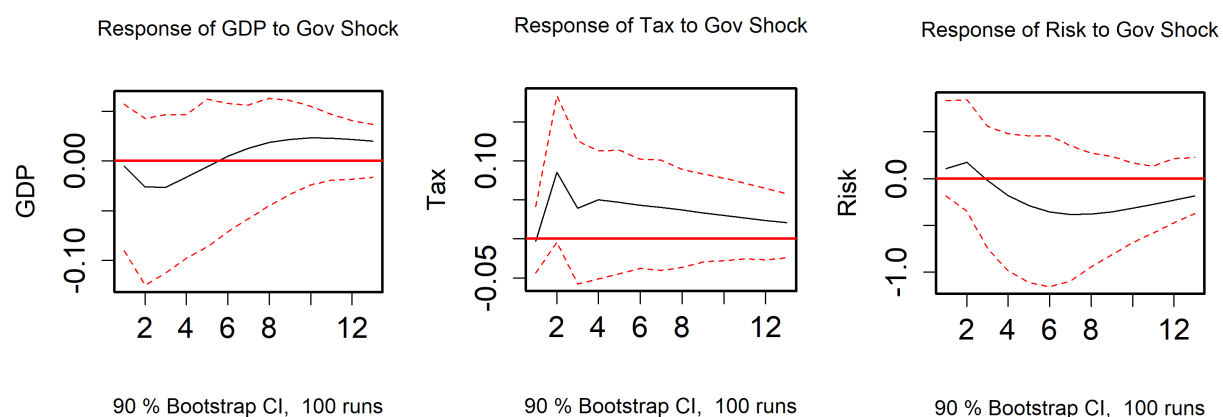


Figure F.18: Effects of Government Spending Shocks - Extended Lag

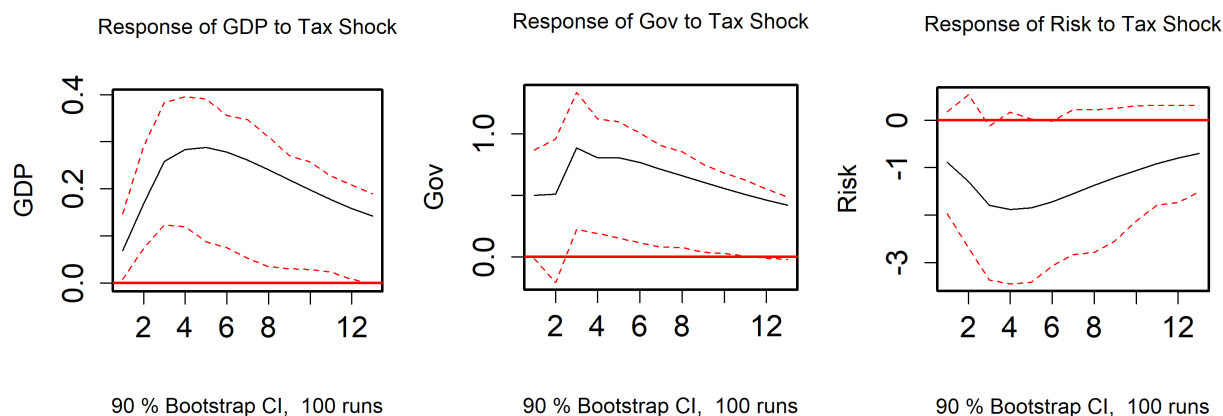


Figure F.19: Effects of Tax Shocks - Extended Lag

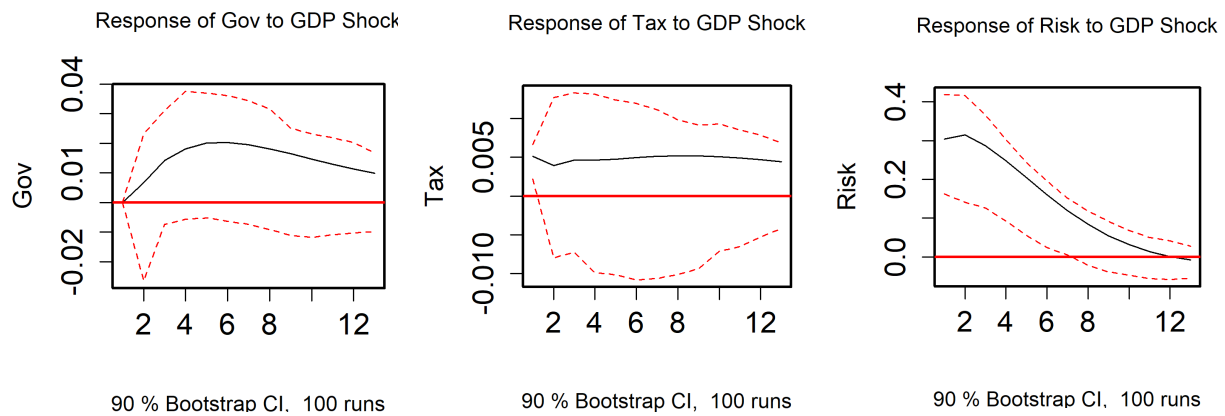


Figure F.20: Effects of GDP Shocks - Extended Lag

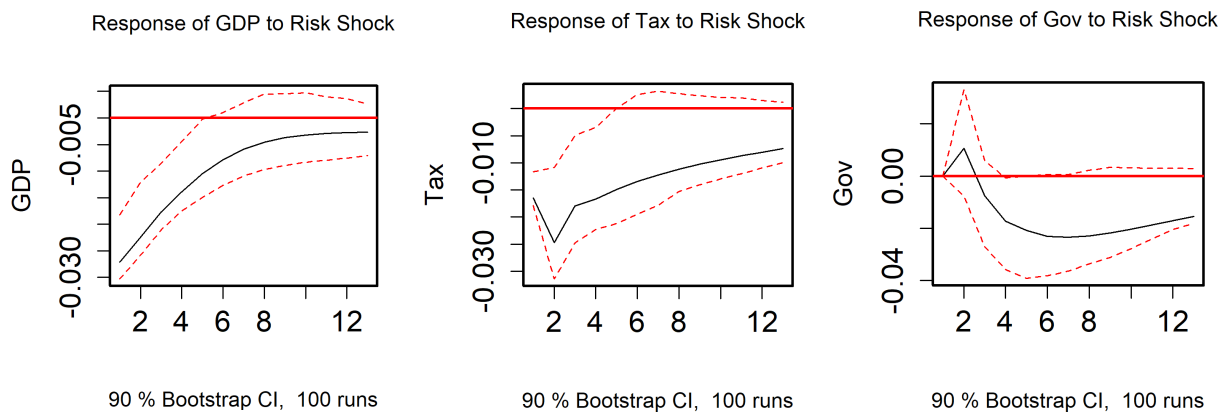


Figure F.21: Effects of Risk Shocks - Extended Lag

	b_{13}	b_{23}	b_{31}	b_{32}	b_{34}	b_{41}	b_{42}	b_{43}	c_{21}
coeff.	-0.6076	0	2.8369	-0.3292	3.2109	3.4979	-0.2949	-38.0812	0.5008

Table F.5: Estimated Contemporaneous Coefficients - Extended Lag

Thesis Code - Fiscal Balance and Economic Growth: A Relationship Conditioned by Sovereign Risk

Aragona Nicolas

2025-11-17

The following code shows each step taken in the Data Visualization and Analysis

Note: the code was done with the help of ChatGPT

Set up

We will download the necessary packages.

```
library(tidyverse)  # Multi-use package
library(readxl)     # Used to read Excel files
library(vars)       # Used to develop the VAR models
library(seasonal)   # Seasonal adjustment (X-13ARIMA-SEATS)
library(stargazer)  # Used to create nice tables of the regression
library(webshot2)   # Used to convert HTML output into image files
```

Now I can load our datasets

```
# now I download the general government fiscal accounts dataset
fisc_arg <- read_excel(
  "C:/HSG/OneDrive - Universität St.Gallen/HSG/Bachelor/Thesis/Data/Final_Data/arg_fin.xlsx"
)
# I delete the superfluous row
fisc_arg <- fisc_arg[-86,] # the other datasets do not have data for Q2 of 2025

# Now I can download the GDP dataset
gdp <- read_excel(
  "C:/HSG/OneDrive - Universität St.Gallen/HSG/Bachelor/Thesis/Data/Final_Data/PIB_Arg.xlsx"
)
# includes real, nominal GDP and deflator

# now we can download the risk dataset
risk_arg <- read_excel(
  "C:/HSG/OneDrive - Universität St.Gallen/HSG/Bachelor/Thesis/Data/Final_Data/Risk_ar.xlsx",
  sheet = 2
)
```

Now I can put all the datasets together

```
df <- left_join(fisc_arg, gdp, by = "Quarter")
df <- left_join(df, risk_arg, by = "Quarter")
```

Now for each fiscal variable I calculate the per capita value

```

# For each variable now I can calculate the per capita value
df$`Gastos Corrientes` <- (df$`Gastos Corrientes` / df$Population) * 1e6
df$`Intereses Netos` <- (df$`Intereses Netos` / df$Population) * 1e6
df$`Gastos de capital` <- (df$`Gastos de capital` / df$Population) * 1e6
df$PIB <- (df$PIB / df$Population) * 1e6
df$`Ingresos Corrientes` <- (df$`Ingresos Corrientes` / df$Population) * 1e6

```

Now I subtract the net interests from the expenses

```

df$tax <- df$`Ingresos Corrientes`
df$spending <- df$`Gastos Corrientes` + df$`Gastos de capital` - df$`Intereses Netos`

```

Now I remove the columns that I do not need

```

df <- df[, c("tax", "spending", "PIB", "Risk", "Deflator")]

```

Now I transform the data to real values

```

df$tax <- (df$tax / df$Deflator) * 100
df$spending <- (df$spending / df$Deflator) * 100
df$PIB <- (df$PIB / df$Deflator) * 100

```

Now I can deseasonize the data

```

T_per_capita <- final(seas(ts(df$tax, frequency = 4, start = c(2004, 1))))
G_per_capita <- final(seas(ts(df$spending, frequency = 4, start = c(2004, 1))))
gdp_per_capita <- final(seas(ts(df$PIB, frequency = 4, start = c(2004, 1))))
Risk <- final(seas(ts(df$Risk, frequency = 4, start = c(2004, 1))))

```

Combine results

```

deseasonalized_df <- data.frame(
  T_per_capita = as.numeric(T_per_capita),
  G_per_capita = as.numeric(G_per_capita),
  gdp_per_capita = as.numeric(gdp_per_capita),
  Risk = as.numeric(Risk)
)
defl <- df$Deflator
df <- deseasonalized_df

```

Now I can put the variables in the dataset

```

df <- df %>%
  rename(
    Tax = T_per_capita,
    Gov = G_per_capita,
    GDP = gdp_per_capita,
    Risk = Risk
  )

```

I make sure now that the variables are in the correct order (and only the ones I want)

```

df <- df[, c("Tax", "Gov", "GDP", "Risk")]

```

Now i can take the logs and create the time series

```
df$Tax<-log(df$Tax)
df$Gov<-log(df$Gov)
df$GDP<-log(df$GDP)
df$Risk<-log(df$Risk)
```

```
ts<-ts(df, start = 2004, frequency = 4)
```

Now I calculate the elasticities in the periods (check if need deaseasonalized data or not)

The output will create pictures that I will use in the thesis

```
Tax_df<-df$Tax
Tax_df<-as.data.frame(Tax_df)
Tax_df$gdp<-df$GDP
```

```
Tax_df <- df %>%
  transmute(
    Tax = (Tax - dplyr::lag(Tax)),
    GDP = (GDP - dplyr::lag(GDP)),
  ) %>%
  drop_na()
```

```
model_taxes<-lm(Tax_df$Tax ~Tax_df$GDP)
summary(model_taxes)
```

```
##
## Call:
## lm(formula = Tax_df$Tax ~ Tax_df$GDP)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.185719 -0.030940  0.004126  0.028045  0.163277
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.001742   0.005982  -0.291  0.77165
## Tax_df$GDP   0.607555   0.201334   3.018  0.00339 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05455 on 82 degrees of freedom
## Multiple R-squared:  0.09995,    Adjusted R-squared:  0.08898
## F-statistic: 9.106 on 1 and 82 DF,  p-value: 0.003393
```

```
elasticity <- coef(model_taxes)["Tax_df$GDP"]
```

```
stargazer(model_taxes,
  type = "html",
  out = "model_taxes_summary.html",
  title = "Regression Results",
  dep.var.labels = "Tax",
  covariate.labels = c("GDP"),
  digits = 4,
```

```
style = "default")
```

```
##
## <table style="text-align:center"><caption><strong>Regression Results</strong></caption>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## <tr><td></td><td colspan="1" style="border-bottom: 1px solid black"></td></tr>
## <tr><td style="text-align:left"></td><td>Tax</td></tr>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## <tr><td style="text-align:left"></td><td>(0.2013)</td></tr>
## <tr><td style="text-align:left"></td><td></td></tr>
## <tr><td style="text-align:left">Constant</td><td>-0.0017</td></tr>
## <tr><td style="text-align:left"></td><td>(0.0060)</td></tr>
## <tr><td style="text-align:left"></td><td></td></tr>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## <tr><td style="text-align:left">R<sup>2</sup></td><td>0.1000</td></tr>
## <tr><td style="text-align:left">Adjusted R<sup>2</sup></td><td>0.0890</td></tr>
## <tr><td style="text-align:left">Residual Std. Error</td><td>0.0545 (df = 82)</td></tr>
## <tr><td style="text-align:left">F Statistic</td><td>9.1062<sup>***</sup> (df = 1; 82)</td></tr>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## </table>
```

```
webshot("model_taxes_summary.html", file = "model_taxes_summary.png",
        vwidth = 320, vheight = 400, zoom = 2)
```

Regression Results

	<i>Dependent variable:</i>
	Tax
GDP	0.6076*** (0.2013)
Constant	-0.0017 (0.0060)
Observations	84
R ²	0.1000
Adjusted R ²	0.0890
Residual Std. Error	0.0545 (df = 82)
F Statistic	9.1062*** (df = 1; 82)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01

```
# I also check if it is actually true that government spending comes with delay
Gov_df<-df$Gov
Gov_df<-as.data.frame(Gov_df)
Gov_df$gdp<-df$GDP
```



```

Gov_df <- df %>%
  transmute(
    Gov = (Gov - dplyr::lag(Gov)),
    GDP = (GDP - dplyr::lag(GDP)),
  ) %>%
  drop_na()

model_gov<-lm(Gov_df$Gov ~Gov_df$GDP) #need to check if this is the correct way to calculate this
summary(model_gov)

##
## Call:
## lm(formula = Gov_df$Gov ~ Gov_df$GDP)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.76009 -0.03272  0.00517  0.03732  0.79669
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.0005054  0.0155739   0.032   0.974
## Gov_df$GDP   0.1791013  0.5241547   0.342   0.733
##
## Residual standard error: 0.142 on 82 degrees of freedom
## Multiple R-squared:  0.001422,    Adjusted R-squared:  -0.01076
## F-statistic: 0.1168 on 1 and 82 DF,  p-value: 0.7335

stargazer(model_gov,
  type = "html",
  out = "model_gov_summary.html",
  title = "Regression Results",
  dep.var.labels = "Gov",
  covariate.labels = c("GDP"),
  digits = 4,
  style = "default")

##
## <table style="text-align:center"><caption><strong>Regression Results</strong></caption>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## <tr><td></td><td colspan="1" style="border-bottom: 1px solid black"></td></tr>
## <tr><td style="text-align:left"></td><td>Gov</td></tr>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## <tr><td style="text-align:left"></td><td>(0.5242)</td></tr>
## <tr><td style="text-align:left"></td><td></td></tr>
## <tr><td style="text-align:left">Constant</td><td>0.0005</td></tr>
## <tr><td style="text-align:left"></td><td>(0.0156)</td></tr>
## <tr><td style="text-align:left"></td><td></td></tr>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## <tr><td style="text-align:left">R<sup>2</sup></td><td>0.0014</td></tr>
## <tr><td style="text-align:left">Adjusted R<sup>2</sup></td><td>-0.0108</td></tr>
## <tr><td style="text-align:left">Residual Std. Error</td><td>0.1420 (df = 82)</td></tr>
## <tr><td style="text-align:left">F Statistic</td><td>0.1168 (df = 1; 82)</td></tr>
## <tr><td colspan="2" style="border-bottom: 1px solid black"></td></tr><tr><td style="text-align:left"
## </table>

```

```
webshot("model_gov_summary.html", file = "model_gov_summary.png", vwidth = 320, vheight = 400, zoom = 2)
```

Regression Results

	<i>Dependent variable:</i>
	Gov
GDP	0.1791 (0.5242)
Constant	0.0005 (0.0156)
Observations	84
R ²	0.0014
Adjusted R ²	-0.0108
Residual Std. Error	0.1420 (df = 82)
F Statistic	0.1168 (df = 1; 82)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01

Now I can create the SVAR model. Before doing this I create the matrices following the identification strategy

```

Amat <- matrix(NA, 4, 4)
Amat[1, ] <- c(1, 0, -elasticity, 0) # Depending on the different specifications
# I can change the elasticity
Amat[2, ] <- c(0, 1, 0, 0) # The elasticity is negative since it,
# after when I expand the equations it will be
Amat[3, ] <- c(NA, NA, 1, NA) # on the other side with a positive sign
Amat[4, ] <- c(NA, NA, NA, 1)

Bmat <- diag(NA, 4,4)
Bmat <- matrix(NA, 4, 4)
Bmat[1, ] <- c(1, 0, 0, 0)
Bmat[2, ] <- c(NA, 1, 0, 0) # here I can also change the c.
Bmat[3, ] <- c(0, 0, 1, 0)
Bmat[4, ] <- c(0, 0, 0, 1)

```

Now I can move on and make the estimation

I start by choosing the optimal lag

```

lagselect <- VARselect(ts, lag.max = 8, type = "both") # has trend and constants
lagselect$criteria

```

```

##              1              2              3              4              5
## AIC(n) -2.038848e+01 -2.028108e+01 -2.012455e+01 -1.995350e+01 -1.991048e+01
## HQ(n)  -2.009627e+01 -1.979406e+01 -1.944273e+01 -1.907687e+01 -1.883905e+01
## SC(n)  -1.965794e+01 -1.906351e+01 -1.841997e+01 -1.776189e+01 -1.723184e+01
## FPE(n)  1.399418e-09  1.565322e-09  1.849782e-09  2.236690e-09  2.406611e-09
##              6              7              8
## AIC(n) -1.996676e+01 -1.996791e+01 -1.981102e+01
## HQ(n)  -1.870052e+01 -1.850687e+01 -1.815517e+01
## SC(n)  -1.680110e+01 -1.631523e+01 -1.567131e+01
## FPE(n)  2.379340e-09  2.532473e-09  3.231998e-09
p <- 1

```

Now I estimate the VAR

```

var_levels <- VAR(ts, p = p, type = "both") # has constant and trend
summary(var_levels)

```

```

##
## VAR Estimation Results:
## =====
## Endogenous variables: Tax, Gov, GDP, Risk
## Deterministic variables: both
## Sample size: 84
## Log Likelihood: 368.757
## Roots of the characteristic polynomial:
## 0.9002 0.7924 0.7924 0.2695
## Call:
## VAR(y = ts, p = p, type = "both")
##
##
## Estimation results for equation Tax:
## =====
## Tax = Tax.l1 + Gov.l1 + GDP.l1 + Risk.l1 + const + trend

```

```

##
##           Estimate Std. Error t value Pr(>|t|)
## Tax.l1    0.6068336  0.1079941   5.619 2.85e-07 ***
## Gov.l1    0.0882646  0.0406594   2.171  0.03299 *
## GDP.l1    0.2739746  0.1499253   1.827  0.07146 .
## Risk.l1   0.0053292  0.0099369   0.536  0.59328
## const    -0.5972095  1.0747283  -0.556  0.58002
## trend    -0.0014355  0.0004432  -3.239  0.00176 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.05369 on 78 degrees of freedom
## Multiple R-Squared:  0.8237, Adjusted R-squared:  0.8124
## F-statistic:  72.9 on 5 and 78 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation Gov:
## =====
## Gov = Tax.l1 + Gov.l1 + GDP.l1 + Risk.l1 + const + trend
##
##           Estimate Std. Error t value Pr(>|t|)
## Tax.l1    0.651721  0.255758   2.548  0.0128 *
## Gov.l1    0.536031  0.096292   5.567 3.54e-07 ***
## GDP.l1   -0.051561  0.355063  -0.145  0.8849
## Risk.l1   0.020183  0.023533   0.858  0.3937
## const    -0.968318  2.545238  -0.380  0.7047
## trend     0.001986  0.001050   1.892  0.0621 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.1272 on 78 degrees of freedom
## Multiple R-Squared:  0.6527, Adjusted R-squared:  0.6304
## F-statistic:  29.31 on 5 and 78 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation GDP:
## =====
## GDP = Tax.l1 + Gov.l1 + GDP.l1 + Risk.l1 + const + trend
##
##           Estimate Std. Error t value Pr(>|t|)
## Tax.l1    0.1437469  0.0543998   2.642  0.00994 **
## Gov.l1   -0.0203910  0.0204814  -0.996  0.32253
## GDP.l1    0.6718518  0.0755219   8.896 1.72e-13 ***
## Risk.l1  -0.0142141  0.0050055  -2.840  0.00576 **
## const     2.4290478  0.5413728   4.487 2.46e-05 ***
## trend     0.0004878  0.0002232   2.185  0.03188 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.02705 on 78 degrees of freedom
## Multiple R-Squared:  0.885, Adjusted R-squared:  0.8776

```

```
## F-statistic: 120 on 5 and 78 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation Risk:
## =====
## Risk = Tax.l1 + Gov.l1 + GDP.l1 + Risk.l1 + const + trend
##
##          Estimate Std. Error t value Pr(>|t|)
## Tax.l1   -0.981733   0.615741  -1.594   0.1149
## Gov.l1   -0.035420   0.231824  -0.153   0.8790
## GDP.l1    1.873744   0.854817   2.192   0.0314 *
## Risk.l1    0.925726   0.056656  16.339 <2e-16 ***
## const   -10.796420   6.127692  -1.762   0.0820 .
## trend    -0.001423   0.002527  -0.563   0.5749
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.3061 on 78 degrees of freedom
## Multiple R-Squared: 0.8419, Adjusted R-squared: 0.8317
## F-statistic: 83.05 on 5 and 78 DF, p-value: < 2.2e-16
##
##
## Covariance matrix of residuals:
##          Tax      Gov      GDP      Risk
## Tax  0.0028826 1.050e-03 6.549e-04 -1.473e-03
## Gov  0.0010500 1.617e-02 2.902e-05 6.604e-05
## GDP  0.0006549 2.902e-05 7.314e-04 -1.027e-03
## Risk -0.0014727 6.604e-05 -1.027e-03 9.371e-02
##
## Correlation matrix of residuals:
##          Tax      Gov      GDP      Risk
## Tax  1.00000 0.153799 0.45099 -0.089608
## Gov  0.15380 1.000000 0.00844 0.001697
## GDP  0.45099 0.008440 1.00000 -0.124035
## Risk -0.08961 0.001697 -0.12404 1.000000
```

Now I do the estimation

```
model <- SVAR(var_levels,
              estmethod = "scoring",
              Amat = Amat,
              Bmat = Bmat,
              max.iter = 5000)
```

```
model$A # these are used to see the contemporaneous coefficients
```

```
##          Tax      Gov      GDP      Risk
## Tax  1.000000 0.0000000 -0.6075548 0.000000
## Gov  0.000000 1.0000000 0.0000000 0.000000
## GDP  3.256230 -0.2265666 1.0000000 3.257668
## Risk 3.681772 -0.1739366 -38.5802680 1.000000
```

```
model$B
```

```
##           Tax Gov GDP Risk
## Tax  1.0000000  0  0  0
## Gov  0.4380022  1  0  0
## GDP  0.0000000  0  1  0
## Risk 0.0000000  0  0  1
```

Now I can produce the Impulse Reaction Functions

```
# Now I can create the IRF plots. This code will directly create a folder where
# It loads the images and a file with all the estimated points
```

```
dir.create("irf_plots", showWarnings = FALSE)
```

```
# Create an empty data frame to store all IRF results
all_irf_values <- data.frame()
```

```
for (imp in colnames(ts)) {
  for (resp in colnames(ts)) {
```

```
    # Run IRF
```

```
    irf_obj <- irf(model,
                  impulse = imp,
                  response = resp,
                  n.ahead = 12,
                  boot = TRUE,
                  runs = 100,
                  ci = 0.90,
                  boot.ci.type = "bca")
```

```
    # Extract numeric values
```

```
    irf_vals <- irf_obj$irf[[imp]][, resp]
    lower_vals <- irf_obj$Lower[[imp]][, resp]
    upper_vals <- irf_obj$Upper[[imp]][, resp]
```

```
    # Combine into one data frame
```

```
    irf_df <- data.frame(
      Horizon = 1:length(irf_vals),
      Impulse = imp,
      Response = resp,
      IRF = irf_vals,
      Lower = lower_vals,
      Upper = upper_vals
    )
```

```
    # Append to master table
```

```
    all_irf_values <- rbind(all_irf_values, irf_df)
```

```
    # Save the plot
```

```
    file_name <- paste0("irf_plots/IRF_", resp, "_to_", imp, "_shock.png")
    png(filename = file_name, width = 1600, height = 1200, res = 300)
```

```
    par(cex.main = 2, font.main = 2,
        cex.lab = 1.2, cex.axis = 1.4,
```

```

        cex.sub = 1.6, font.sub = 2, lwd = 2)

    plot(irf_obj,
         main = paste("Response of", resp, "to", imp, "Shock"))

    dev.off()
  }
}

# Save all numeric IRFs to one CSV
write.csv(all_irf_values, "irf_plots/All_IRF_values.csv", row.names = FALSE)

cat(" All IRF values saved to: irf_plots/All_IRF_values.csv\n")

## All IRF values saved to: irf_plots/All_IRF_values.csv

```

Additional Visualization

```

ts <- exp(ts)
ts <- cbind(ts, defl)
colnames(ts) <- c("Tax", "Gov", "GDP", "Risk", "Deflator")

# This creates a plot for each variable in the model and directly creates an image
png("variables_plot.png", width = 2500, height = 2000, res = 300)

par(
  mfcol = c(3, 2),
  mar = c(3.5, 4, 2, 1),
  oma = c(1, 1, 3, 1),
  mgp = c(2, 0.7, 0),
  cex = 0.9
)

plot(ts[, "Tax"], main = "Tax", xlab = "Year", ylab = "Tax", type = "l")
plot(ts[, "Risk"], main = "Risk", xlab = "Year", ylab = "Risk", type = "l")
plot(ts[, "Gov"], main = "Gov", xlab = "Year", ylab = "Gov", type = "l")
plot(ts[, "Deflator"], main = "Deflator", xlab = "Year", ylab = "Deflator", type = "l")
plot(ts[, "GDP"], main = "GDP", xlab = "Year", ylab = "GDP", type = "l")

mtext("Macroeconomic Indicators Over Time",
      outer = TRUE, cex = 1.4, font = 2, line = 0.5)

dev.off()

## pdf
## 2

# Now I can plot the graph used for poverty

pobreza <- read_excel(
  "C:/HSG/OneDrive - Universität St.Gallen/HSG/Bachelor/Thesis/Data/Final_Data/pobreza.xls",
  sheet = 2

```

```

)

pobreza$...1 <- as.Date(pobreza$...1, format = "%d/%m/%Y")
df_pob<-pobreza[,c(1,2,3)]
df_indig<-pobreza[,c(1,4,5)]

# Compute the global range
y_range <- range(df_pob$`Pob. Hog`, df_pob$`Pob. Pers`)

png("poverty.png", width = 900, height = 600, res =150)

plot(df_pob$...1, df_pob$`Pob. Hog`, type = "l", col = "blue",
      xlab = "Date", ylab = "Population (as %)",
      main = "Poverty over time in Argentina",
      ylim = y_range)

lines(df_pob$...1, df_pob$`Pob. Pers`, col = "red")

legend("topleft", legend = c("Households", "People"),
      col = c("blue", "red"), lty = 1, bty = "n")
dev.off()

## pdf
## 2

# Calculate combined y-axis range so all values fit
y_range <- range(df_indig$`Indig. Hog`, df_indig$`Indig. Pers`)

png("ex_poverty.png", width = 900, height = 600, res =150)

# Plot the first variable
plot(df_indig$...1, df_indig$`Indig. Hog`, type = "l", col = "blue",
      xlab = "Date", ylab = "Population (as %)",
      main = "Extreme Poverty over Time",
      ylim = y_range)

# Add the second variable
lines(df_indig$...1, df_indig$`Indig. Pers`, col = "red")

# Add a legend
legend("topleft",
      legend = c("Households", "People"),
      col = c("blue", "red"),
      lty = 1, bty = "n")

dev.off()

## pdf
## 2

# Now I can create the the plot for the debt

debt <- read_excel(
  "C:/HSG/OneDrive - Universität St.Gallen/HSG/Bachelor/Thesis/Data/Final_Data/debt_arg.xls"

```



```

)

# Reshape data
debt_long <- debt %>%
  pivot_longer(
    cols = starts_with("19") | starts_with("20"),
    names_to = "Year",
    values_to = "DebtPercentGDP"
  )

debt_long <- debt_long[,-1] # remove first column, I do not need it

# Convert Year to numeric
debt_long$Year <- as.numeric(debt_long$Year)

png("debt.png", width = 900, height = 600, res = 150)
# Plot
# Plot your main graph
plot(
  debt_long$Year,
  debt_long$DebtPercentGDP,
  type = "l",
  xlab = "Year",
  ylab = "Debt (% of GDP)",
  lwd = 2
)

dev.off()

## pdf
## 2

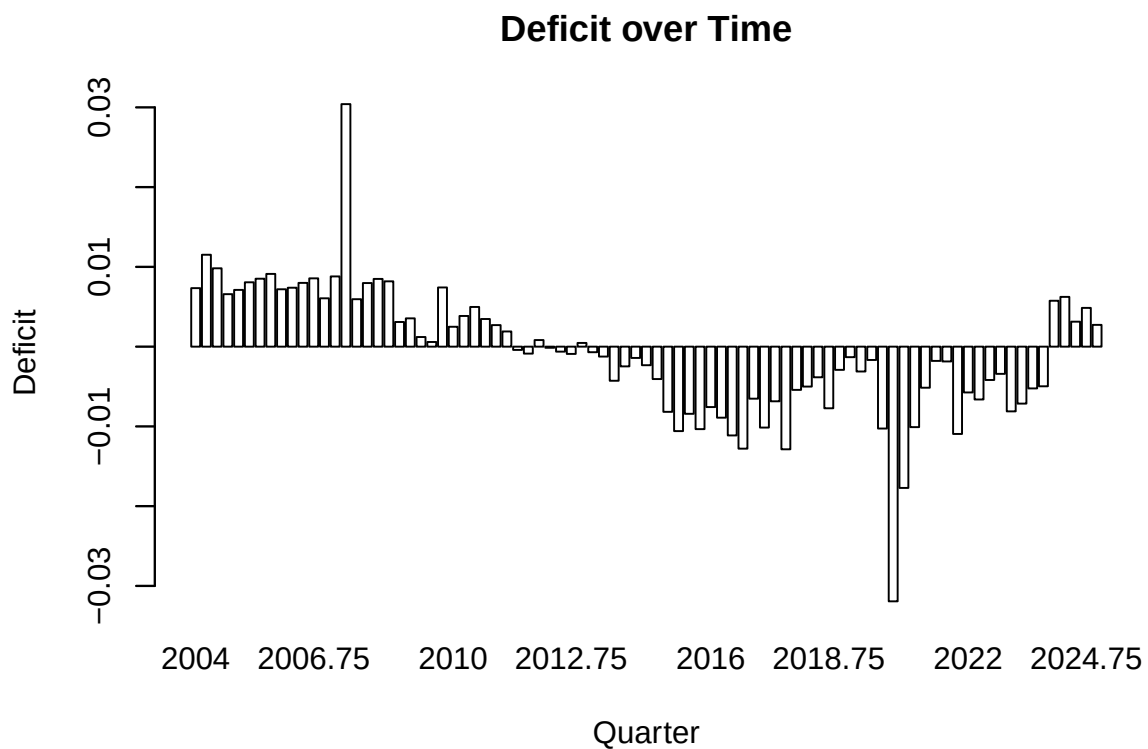
# Now I want to plot the fiscal balance of Argentina

df<-as.data.frame(ts)
df$quarter<-time(ts)

df$deficit<-(df$Tax-df$Gov)/df$GDP
df<-df[, c("quarter", "deficit")]

barplot(df$deficit,
  names.arg = df$quarter,
  xlab = "Quarter",
  ylab = "Deficit",
  main = "Deficit over Time",
  col = "white",
  border = "black",
  lwd = 1.2)

```



```
# Now I can plot the Inflation plot

inflation <- read_excel(
  "C:/HSG/OneDrive - Universität St.Gallen/HSG/Bachelor/Thesis/Data/Final_Data/inflation_ARG.xls",
  sheet = 2
)

inflation$Date<-as.Date(inflation$Date)

png("inflation.png", width = 900, height = 600, res =150)

plot(
  inflation$Date,
  inflation$Value,
  type = "l",
  xlab = "Year",
  ylab = "Monthly Inflation Rate (%)",
  lwd = 2
)

abline(v = as.Date("2019-12-31"), col = "black", lty = 2, lwd = 2)
abline(v = as.Date("2023-12-31"), col = "black", lty = 2, lwd = 2)

par(xpd = TRUE)

ylim <- par("usr")[4]
```

```

text(as.Date("2018-01-01"), ylim + 1, "Mauricio Macri", cex = 0.7)
text(as.Date("2022-01-01"), ylim + 1, "Alberto Fernández", cex = 0.7)
text(as.Date("2024-12-01"), ylim + 1, "Javier Milei", cex = 0.7)

dev.off()

## pdf
## 2

png("gdp.png", width = 3000, height = 1700, res = 300)

# Plot the GDP time series
plot(ts[, "GDP"],
     main = " Real GDP per capita in Argentina",
     xlab = "Year",
     ylab = "GDP (in real pesos, base = 2004)",
     type = "l",
     lwd = 2)

# Add vertical lines for each presidency start
abline(v = 2003 + 5/12, col = "black", lty = 2) # May 2003 - Néstor Kirchner
abline(v = 2007 + 11/12, col = "black", lty = 2) # Dec 2007 - Cristina Kirchner
abline(v = 2015 + 11/12, col = "black", lty = 2) # Dec 2015 - Macri
abline(v = 2019 + 11/12, col = "black", lty = 2) # Dec 2019 - Alberto Fernández
abline(v = 2023 + 11/12, col = "black", lty = 2) # Dec 2023 - Javier Milei

# Define x positions (at each line) and labels
x_positions <- c(2003 + 5/12, 2007 + 11/12, 2015 + 11/12, 2019 + 11/12, 2023 + 11/12)
presidents <- c("N. Kirchner", "C. Kirchner", "M. Macri",
               "Fernández", "Milei")

# Add the names horizontally above the plot
text(x = x_positions + 1.5,
     y = par("usr")[4] + 0.05 * diff(par("usr")[3:4]), # slightly above the top y-limit
     labels = presidents,
     srt = 0, # keep text horizontal
     adj = 0.5, # center align
     xpd = TRUE, # allow drawing outside plot area
     cex = 0.8, # font size
     font = 1,
     lwd = 2)

dev.off()

## pdf
## 2

```

Declaration of Authorship

I hereby declare,

- that I have written this thesis independently,
- that I have written the thesis using only the aids specified in the index;
- that all parts of the thesis produced with the help of aids have been declared;
- that I have handled both input and output responsibly when using AI. I confirm that I have therefore only read in public data or data released with consent and that I have checked, declared and comprehensibly referenced all results and/or other forms of AI assistance in the required form and that I am aware that I am responsible if incorrect content, violations of data protection law, copyright law or scientific misconduct (e.g. plagiarism) have also occurred unintentionally;
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List of Aids

The following lists reflects to the aids that were used in this thesis.

Aid / Tool	Purpose
Gemini	Check grammar and spelling errors and also improving the flow of the thesis
R	Used to generate the graphs and the model
Chat GPT	Used in the reformulation of sentences previously written by me. Additionally, it helped me to write the code of my model and graphs on R. Used also to generate the LaTeX code for this thesis
Scribbr	Used as a citation aid and as a platform to organize my sources and eased the process of integrating them into my LaTeX document

Date

Signature